8. A charged particle of mass m having kinetic energy K passes undeflected through a region with electric field $\vec{E}$ and magnetic field $\vec{B}$ acting perpendicular to each other. The mass m of the particle will be :

(A) $\dfrac{KB^2}{2E^2}$

(B) $\dfrac{2KB^2}{E^2}$

(C) $\dfrac{2KE^2}{B^2}$

(D) $\dfrac{KE^2}{2B^2}$

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(D) $\frac{KE^2}{2B^2}$

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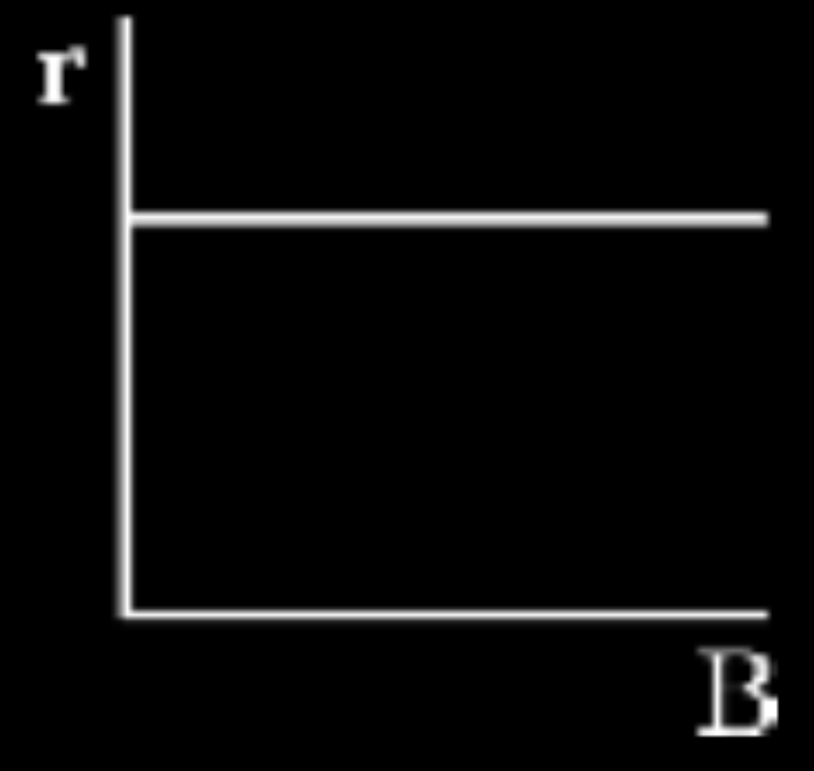
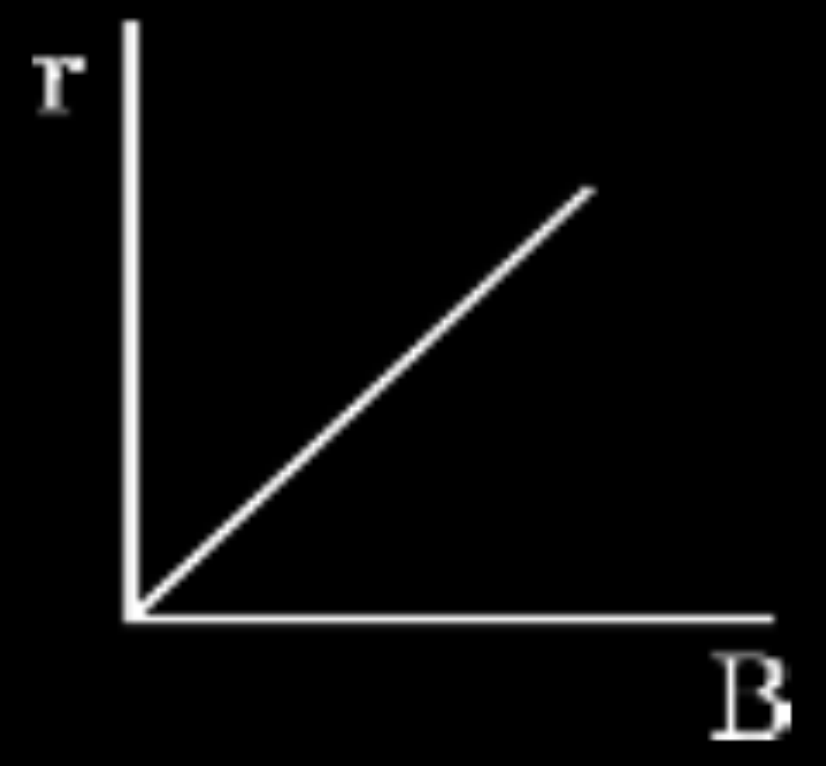
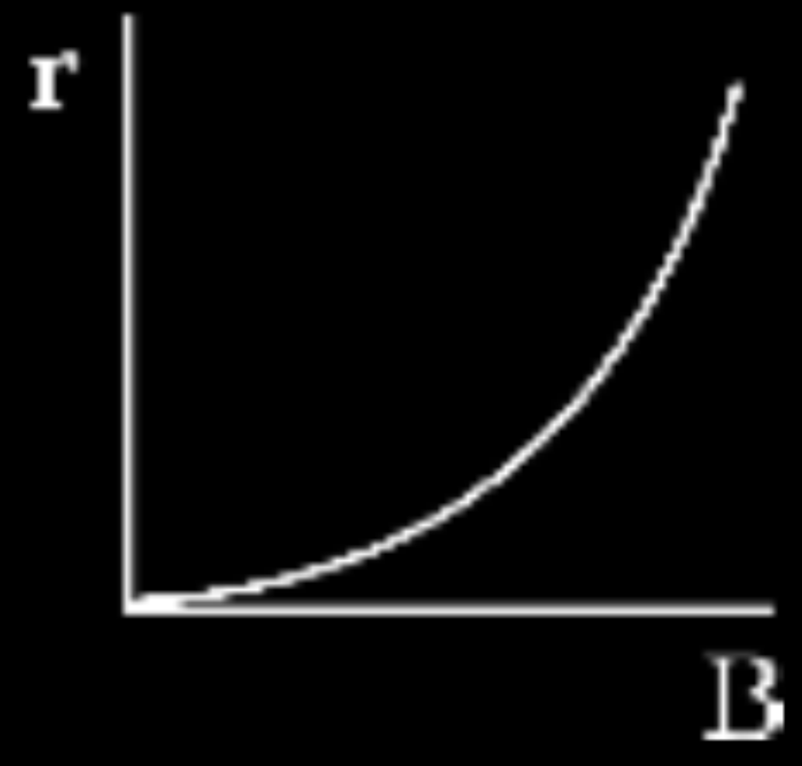
(B) $\frac{2KB^2}{E^2}$

5. A charged particle is moving in a uniform magnetic field $\vec{B}$ with a constant speed v in a circular path of radius r. Which of the following graphs represents the variation of radius of the circle, with the magnitude of magnetic field $\vec{B}$? CBSE 2026

(A) 

(B) 

(C) 

(D) 

5. A charged particle is moving in a uniform magnetic field $\vec{B}$ with a constant speed v in a circular path of radius r. Which of the following graphs represents the variation of radius of the circle, with the magnitude of magnetic field $\vec{B}$?

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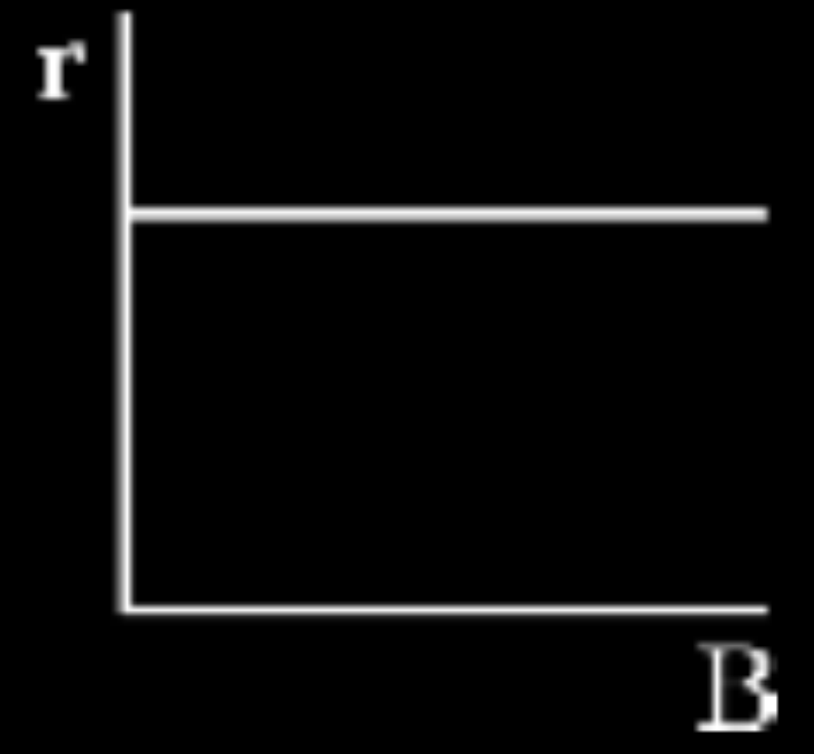
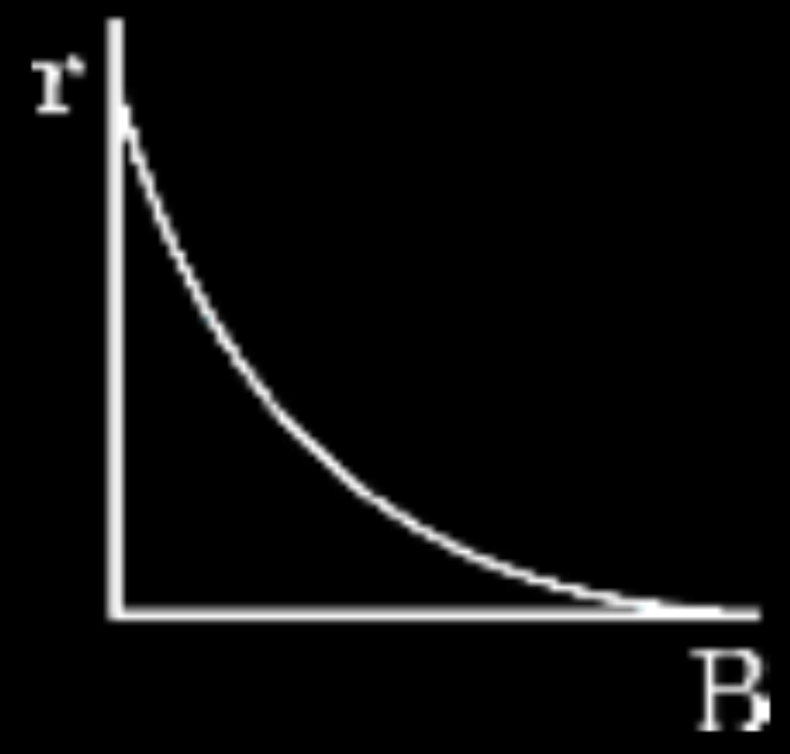
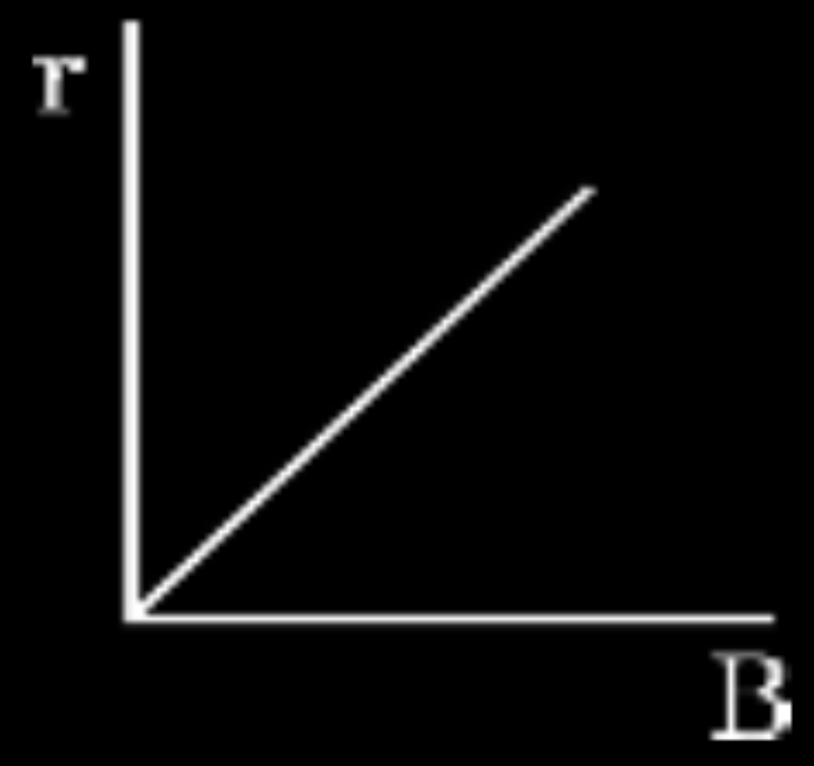
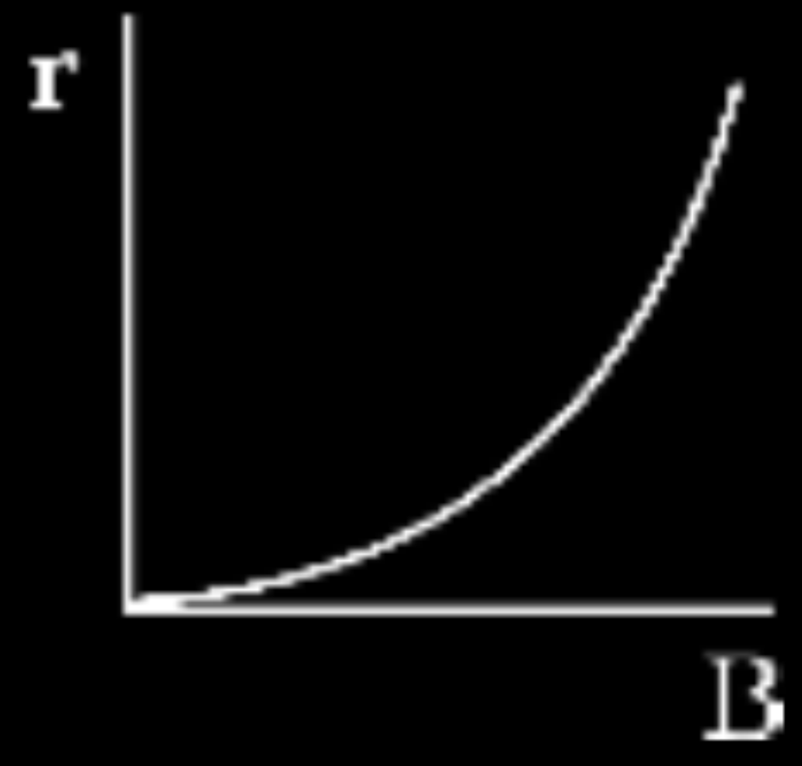
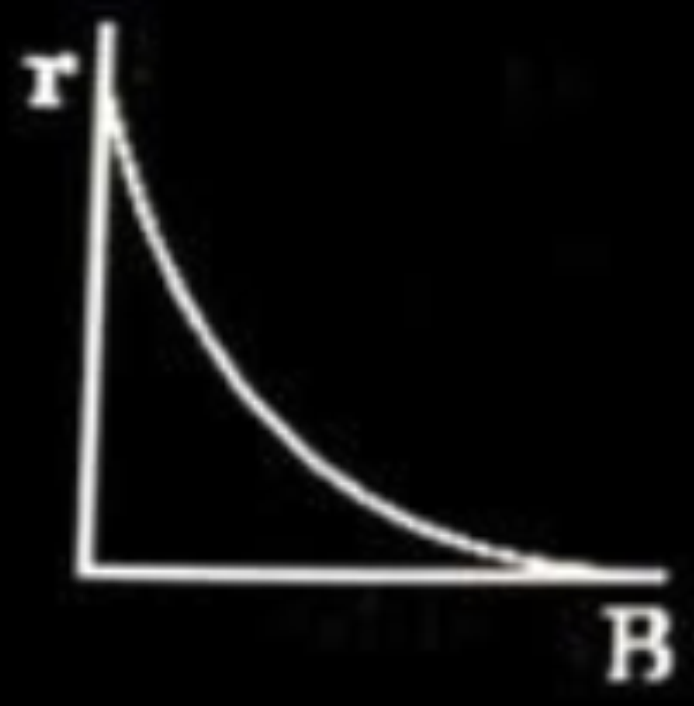
(A) 

(B) 

(C) 

(D) 

(B)



2. Two particles of masses m_1 and m_2 having charges q_1 and q_2 respectively are projected with the same velocity in a region of uniform magnetic field $\vec{B}$ pointing vertically upward. If they describe circular paths as shown in the figure, one may conclude that :

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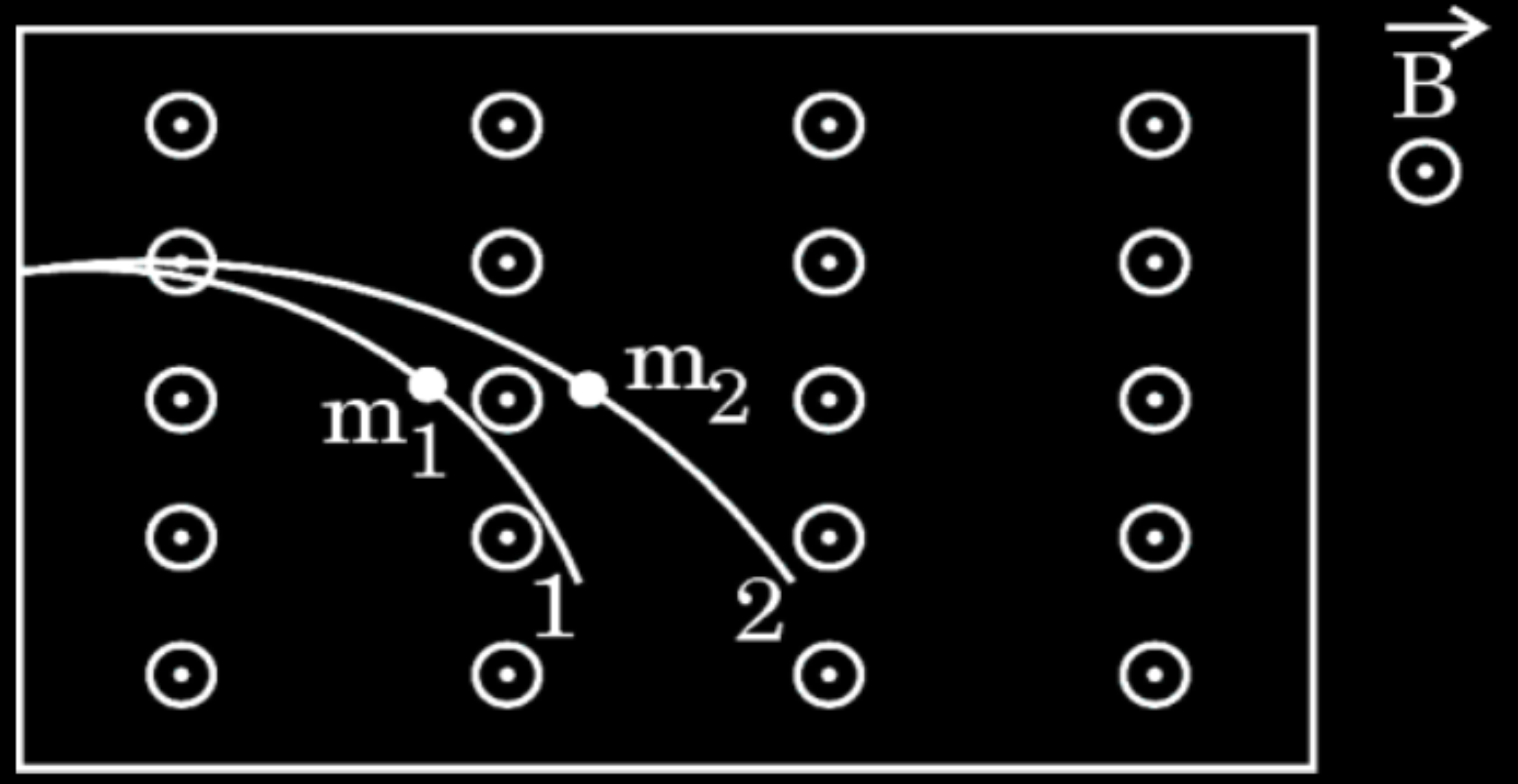

(A) $\frac{m_1}{m_2} > \frac{q_1}{q_2}$

(B) $\frac{m_1}{m_2} > \frac{q_2}{q_1}$

(C) $\frac{m_1}{m_2} < \frac{q_1}{q_2}$

(D) $\frac{m_1}{m_2} < \frac{q_2}{q_1}$

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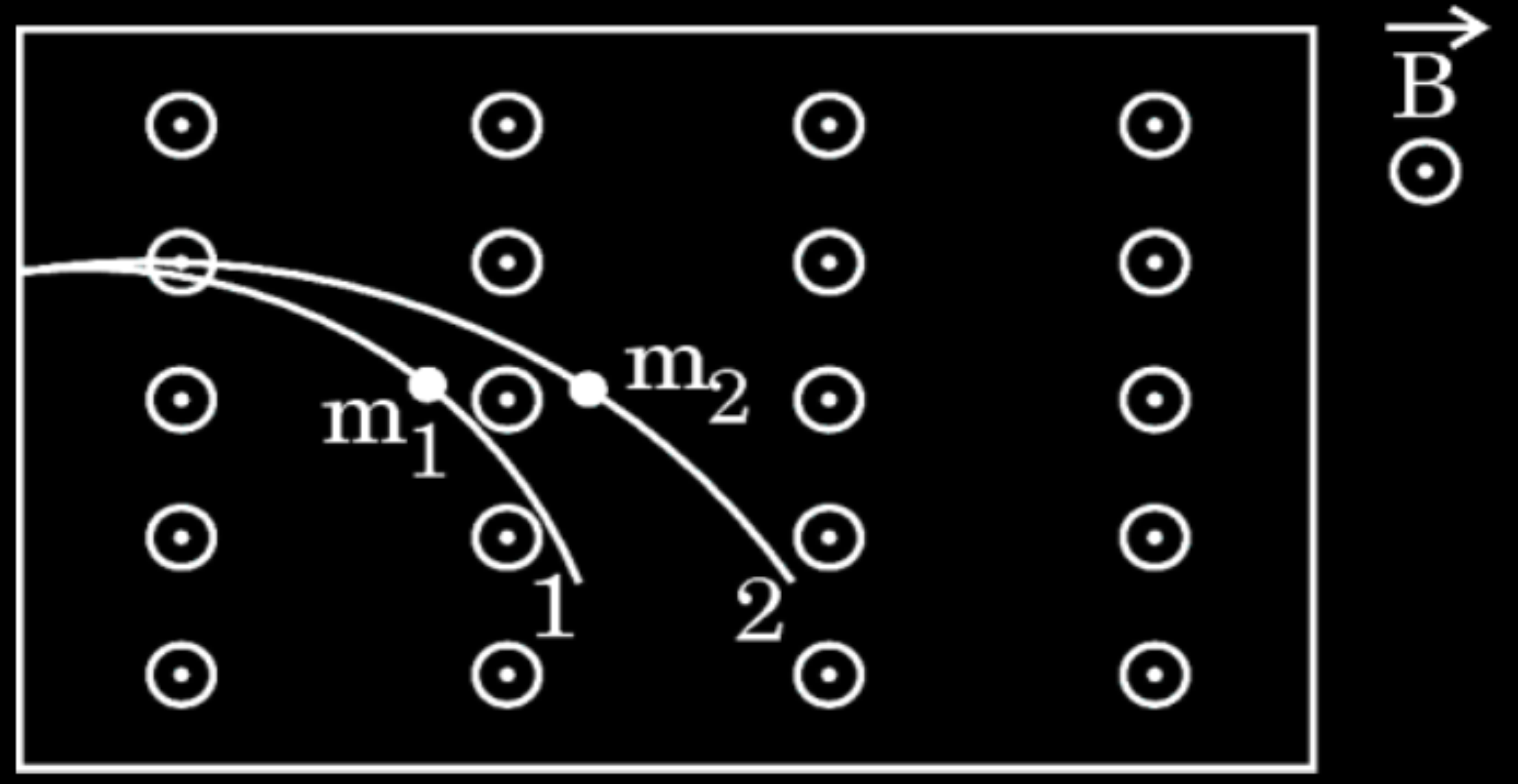

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(D) $\frac{m_1}{m_2} < \frac{q_2}{q_1}$

(C) $\frac{m_1}{m_2} < \frac{q_1}{q_2}$

(b) An alpha particle (mass $6 \cdot 4 \times 10^{-27}$ kg and charge $3 \cdot 2 \times 10^{-19}$ C) having $8 \cdot 0$ MeV energy, enters a region of a uniform magnetic field of $0 \cdot 5$ T. If the field is directed perpendicular to the velocity of the particle, find the radius of the circular path described by the particle. Mention the condition under which the particle in this region (i) describes a helical path, and (ii) goes straight undeviated. *3*

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$K.E = \frac{q^2B^2r^2}{2m}$	½
$r = \frac{\sqrt{2 \times K.E. \times m}}{qB}$	
$= \frac{\sqrt{2 \times 8 \times 1.6 \times 10^{-19} \times 10^6 \times 6.4 \times 10^{-27}}}{3.2 \times 10^{-19} \times 0.5}$	½
$= \frac{12.8 \times 10^{-20}}{1.6 \times 10^{-19}}$	½
$= 8 \times 10^{-1}$ m $= 80$ cm	½
(i) Particle enters the magnetic field at angle $0^\circ < \theta < 90^\circ$	½
(Alternatively) If a component of velocity is parallel or antiparallel to the magnetic field and another component is perpendicular to the magnetic field.	
(ii) Particle moves parallel or antiparallel to the magnetic field.	½

30. A charged particle +q in an electric field $\vec{E}$ experiences a force in the direction of the electric field. As a result, its kinetic energy changes. Similarly, the charged particle also experiences a force when it moves in a magnetic field $\vec{B}$. But this magnetic force is perpendicular to both velocity $\vec{v}$ of the charged particle and the magnetic field $\vec{B}$, so it cannot change the kinetic energy of the charged particle. Consider two charged particles 1 and 2 of masses m and $\frac{m}{2}$ having charges $-q$ and $+2q$ respectively. They are accelerated from rest through the same potential difference V and acquire kinetic energy K_1 and K_2. Then they enter in a region of uniform magnetic field $\vec{B}$ perpendicular to their velocities. CBSE 2026

(i) The ratio of their kinetic energies $\left(\frac{K_1}{K_2}\right)$ is : *1*

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) 4 (D) 1

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(C) 4 (D) 1

(A) $\frac{1}{2}$

(ii) The ratio of the radii of the circular paths described by them $\left(\frac{r_1}{r_2}\right)$ is : *1*

(A) $\frac{1}{\sqrt{2}}$

(B) $\sqrt{2}$

(C) $\frac{1}{2}$

(D) 2

(ii) The ratio of the radii of the circular paths described by them $\left(\frac{r_1}{r_2}\right)$ is : *1*

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$

(C) $\frac{1}{2}$ (D) 2

(D) 2

(iii) Suppose particles 1 and 2 enter the magnetic field $\vec{B} = B_0 \hat{k}$ with velocities $\vec{v}_1 = v_1 \hat{i}$ and $\vec{v}_2 = v_2 \hat{i}$. Then : *1*

(A) both particles revolve clockwise

(B) both particles revolve anticlockwise

(C) particle 1 revolves clockwise while particle 2 revolves anticlockwise

(D) particle 1 revolves anticlockwise while particle 2 revolves clockwise

(iii) Suppose particles 1 and 2 enter the magnetic field $\vec{B} = B_0 \hat{k}$ with velocities $\vec{v}_1 = v_1 \hat{i}$ and $\vec{v}_2 = v_2 \hat{i}$. Then : *1*

(A) both particles revolve clockwise

(B) both particles revolve anticlockwise

(C) particle 1 revolves clockwise while particle 2 revolves anticlockwise

(D) particle 1 revolves anticlockwise while particle 2 revolves clockwise

(D) Particle 1 revolves anticlockwise while particle 2 revolves clockwise

(iv) (a) If period of revolution for particle 1 is 4 s, then for particle 2, the period will be : *1*

(A) 1 s

(B) 2 s

(C) 4 s

(D) 8 s

OR

(b) If the value of momentum for particles 1 and 2 are p_1 and p_2, then : *1*

(A) $p_1 = \frac{p_2}{2}$

(B) $p_1 = p_2$

(C) $p_1 = 2p_2$

(D) $p_1 = 4p_2$

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(B) $p_1 = p_2$

(C) $p_1 = 2p_2$

(D) $p_1 = 4p_2$

(a) (B) $4\mu C$

OR

(b) (C) remain the same

30. A charged particle is moving horizontally with a velocity $\vec{v}$. If a uniform magnetic field $\vec{B}$ directly vertically up is set-up in the region, the particle moves in a circle of radius R in anti-clockwise direction. **4 × 1 = 4**

(i) The charged particle exhibiting this motion is

(A) electron (B) proton

(C) alpha-particle (D) neutron

(ii) The direction of the magnetic force acting on the moving particle is given by –

(A) Biot-Savart's law (B) Lenz's law

(C) Ampere's circuital law (D) Right hand rule

(iii) (a) If the velocity of the particle is doubled, the radius of the circle described will be

(A) $\frac{R}{4}$ (B) $\frac{R}{2}$

(C) 2R (D) R

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(i) (A) electron

(ii) (D) Right-hand Rule

(iii) (a) (C) 2R

(b) A particle having charge 1 μC is moving with a velocity of $5 \times 10^6 \ ms^{-1}$ in a uniform magnetic field of 1T. If the angle between the velocity $(\vec{v})$ and the magnetic field $(\vec{B})$ is 30°, the magnitude of the force on the particle will be –

(A) 25 N

(B) 5 N

(C) 2.5 N

(D) 1.2 N

(b) A particle having charge 1 μC is moving with a velocity of 5×10^6 ms^{-1} in a uniform magnetic field of 1T. If the angle between the velocity $(\vec{v})$ and the magnetic field $(\vec{B})$ is 30°, the magnitude of the force on the particle will be –

(A) 25 N (B) 5 N

(C) 2.5 N (D) 1.2 N

(C) 2.5N

(iv) The frequency of revolution of a particle of mass m and charge q, moving with speed v along a circular path of radius R, in magnetic field $\vec{B}$ (perpendicular to $\vec{v}$) will be

(A) $\frac{mv}{qB}$ (B) $\frac{qB}{2\pi m}$

(C) $\frac{qB}{2\pi R}$ (D) $\frac{\pi m}{2qB}$

(iv) The frequency of revolution of a particle of mass m and charge q, moving with speed v along a circular path of radius R, in magnetic field $\vec{B}$ (perpendicular to $\vec{v}$) will be

(A) $\frac{mv}{qB}$

(B) $\frac{qB}{2\pi\, m}$

(C) $\frac{qB}{2\pi\, R}$

(D) $\frac{\pi m}{2q\, B}$

(B) $\frac{qB}{2\pi m}$

3. A charged particle at rest is subjected to an external magnetic field $\vec{B}$. If no other force acts on the particle, then it will :

(a) accelerate along $\vec{B}$

(b) remain at rest

(c) move in a circular path

(d) move with a constant speed along $\vec{B}$

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(a) accelerate along $\vec{B}$

(b) remain at rest

(c) move in a circular path

(d) move with a constant speed along $\vec{B}$

(b) remain at rest

(b) (i) A particle of mass m and charge q, at the origin moves with a velocity $\vec{v}$ in xy-plane making an angle θ with x-axis. It is subjected to a uniform magnetic field $\vec{B}$ along x-axis. Justify that the particle will move in a helical path. Hence, obtain expression for the radius of the helix.

(ii) A long straight wire kept horizontally carries a current of 4 A from west to east direction. Find the direction and magnitude of magnetic field $\vec{B}$ at a point 20 cm below the wire. *5*

(i) Consider two long parallel conductors P and Q, distance r apart carrying currents I_1 and I_2 respectively.

magnetic field of P at a point in the position of Q

$$B_1 = \frac{\mu_o I_1}{2\pi r} \text{\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_(i)}$$

Magnetic force on Q at length l

$$F_{QP} = B_1 I_2 l = \frac{\mu_o I_1 I_2}{2\pi r} l \text{\_\_\_\_\_\_\_\_\_\_\_\_(ii)}$$ ½

$$f_2 = \frac{F_{QP}}{l} = \frac{\mu_o I_1 I_2}{2\pi r} \text{\_\_\_\_\_\_\_\_\_\_\_\_(iii)}$$ ½

Similarly force per unit length of P due to magnetic field of Q

$$f_1 = \frac{F_{PQ}}{l} = \frac{\mu_o I_1 I_2}{2\pi r} \text{\_\_\_\_\_\_\_\_\_\_\_\_(iv)}$$ ½

From eq. (ii) and (iv)

$$f_1 = f_2 = f = \frac{\mu_o I_1 I_2}{2\pi r} \text{\_\_\_\_\_\_\_\_\_\_(v)}$$

Definition of one ampere

One ampere is the value of the steady current which when maintained in each of the two very long straight parallel conductors of negligible cross section, placed one meter apart in vaccum, would produce on each of these conductors a force equal to 2×10^{-7} newton per meter of length. 1

The force between the two conductors will be attractive when they carry current in the same direction. ½

(ii) Magnetic field at the centre of the current carrying coil

$$B = N\frac{\mu_0 I}{2a}$$ ½

a= 6.28 cm N = 50 I = 4A

$$B = \frac{50\times4\pi\times10^{-7}\times4}{2\times6.28\times10^{-2}}$$ ½

$$= 4\times10^{-3}\,\text{T}$$ ½

When radius of the coil is halved i.e. $\frac{a}{2}$ then magnetic field becomes two times. $B = 8\times10^{-3}$ T ½

23. A particle of charge q and mass m starts moving with uniform velocity $v_0 \hat{i}$. Specify the direction of magnetic field which should be set up in the region so that the particle moves (a) straight undeviated, and (b) in a circle. Justify your answers. *2*

Specification of the direction of magnetic field for particle to move	
(a) straight undeviated	1
(b) in a circle	1

(a) For straight undeviated path, magnetic field should be set up parallel or antiparallel to velocity vector	½
$\vec{F} = q(\vec{v} \times \vec{B})$	½
Altrenatively: angle between $\vec{v}$ *and* $\vec{B}$, $\theta = 0^\circ$, or $\theta = 180^\circ$	
$\theta = 0^\circ$, or $\theta = 180^\circ$	
(b) If charge particle moves in circle magnetic field should be setup	½
perpendicular to the velocity vector.	½
$\vec{F} = q(\vec{v} \times \vec{B}) \qquad \vec{v} \perp \vec{B}$	

2. An electron enters a uniform magnetic field with speed v. It describes a semicircular path and comes out of the field. The final speed of the electron is :

(a) Zero (b) v (c) $\frac{v}{2}$ (d) 2v

2. An electron enters a uniform magnetic field with speed v. It describes a semicircular path and comes out of the field. The final speed of the electron is :

(a) Zero (b) v (c) $\frac{v}{2}$ (d) 2v

(b) v

5. A particle of mass m and charge q moving with a uniform velocity $\vec{v} = v_{0x}\,\hat{i} + v_{0y}\,\hat{j}$ enters a region with a magnetic field $\vec{B} = B_0\,\hat{j}$. After some time, an electric field $\vec{E} = E_0\,\hat{j}$ is also switched on in the region. The resulting path described by the particle will be :

(a) a circle in x-z plane

(b) a parabola in x-y plane

(c) a helix with constant pitch

(d) a helix with increasing pitch

5. A particle of mass m and charge q moving with a uniform velocity $\vec{v} = v_{0x}\hat{i} + v_{0y}\hat{j}$ enters a region with a magnetic field $\vec{B} = B_0\hat{j}$. After some time, an electric field $\vec{E} = E_0\hat{j}$ is also switched on in the region. The resulting path described by the particle will be :

(a) a circle in x-z plane

(b) a parabola in x-y plane

(c) a helix with constant pitch

(d) a helix with increasing pitch

(d) a helix with increasing pitch.

21. An alpha particle is projected with velocity $\vec{v} = (3{\cdot}0 \times 10^5 \text{ m/s})\ \hat{i}$ into a region in which magnetic field $\vec{B} = [(0{\cdot}4 \text{ T})\ \hat{i} + (0{\cdot}3 \text{ T})\ \hat{j}\ ]$ exists. Calculate the acceleration of the particle in the region. $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along x, y and z axis respectively and charge to mass ratio for alpha particle is $4{\cdot}8 \times 10^7$ C/kg. *2*

$\vec{F}=q(\vec{v}\times\vec{B})$	½
$= q(3\times10^5\hat{i}\times(0.4\hat{i}+0.3j))N$	
$\vec{F} = q(0.9\times10^5 k)N$	½
$\vec{F} = \mathrm{m}\vec{a} = q(0.9\times10^5 k)N$	
$\vec{a} = \frac{q}{m}(0.9\times10^5 k)ms^{-2}$	½
$= 4.8\times10^7 \times 0.9\times10^5\ k\ \mathrm{ms}^{-2}$ $= 4.32\times10^{12}\ k\ \mathrm{ms}^{-2}$	½
Note: Deduct ½ mark if a student does not mention the direction of acceleration.	

(b) A long straight conductor kept along X' X axis, carries a steady current I along + x direction. At an instant t, a particle of mass m and charge q at point (x, y) moves with a velocity $\vec{v}$ along + y direction. Find the magnitude and direction of the force on the particle due to the conductor. **2**

(b) Calculation of magnitude of force	(1½)
Direction	(½)

$\vec{F} = q(\vec{V} \times \vec{B})$	½
$B = \dfrac{\mu_o I}{2\pi y}$ and $\theta = 90^0$	½
$F = \dfrac{\mu_o qvI}{2\pi y}$	½
Force is along + X axis	½
Alternatively	
$\vec{F} = \dfrac{\mu_o qvI}{2\pi y}\hat{i}$	

23. (a) Write the expression for the Lorentz force on a particle of charge q moving with a velocity $\vec{v}$ in a magnetic field $\vec{B}$. When is the magnitude of this force maximum ? Show that no work is done by this force on the particle during its motion from a point $\vec{r}_1$ to point $\vec{r}_2$. **2**

Expression for Lorentz force	(½)
Condition for maximum magnitude of force	(½)
Showing no work done in moving the charge from point $\vec{r}_1$ *to* $\vec{r}_2$	(1)

$\vec{F}_m = q\left(\vec{v} \times \vec{B}\right)$ $F_m = q\,v\,B\sin\theta \quad (\because \vec{v} \perp \vec{B})$	½
Force is maximum for $\theta = 90^0$ As magnetic force always acts in a direction perpendicular to the velocity vector, hence no work is done by this force on the particle during motion. **Alternatively** $W = \vec{F}.(\vec{r_2} - \vec{r_1})$ $= F\left\|\vec{r_2} - \vec{r_1}\right\|\cos 90^\circ$ $= 0$	½ 1

32. (a) (i) A charged particle with speed v_0 enters a region in which a strong and non-uniform magnetic field exists everywhere. It comes out of the region following a complicated trajectory without suffering any collision in the region. Would its final speed v be equal to its initial speed v_0 ? Justify your answer.

(ii) In a region, a uniform magnetic field of 6×10^{-4} T is maintained. An electron enters the field with a speed of 3×10^{6} ms^{-1} normal to the field. Find radius of its path. Also calculate its energy in eV. *5*

(i) Answer	1
Justification	1
(ii) Calculation of radius	1 ½
Calculation of energy in eV	1 ½

(i) Yes, the final speed v would be equal to its initial speed v_o.	1
Because the magnetic fore is perpendicular to the direction of motion of the charged particle.	1
(ii) $r = \frac{m\text{v}}{qB}$	½
$= \frac{9.1\times10^{-31}\times3\times10^{6}}{1.6\times10^{-19}\times6\times10^{-4}}$ = 2.8 cm	1
K.E= $\frac{1}{2}m\text{v}^2$ $= \frac{\frac{1}{2}\times9.1\times10^{-31}\times(3\times10^{6})^2}{1.6\times10^{-19}}\ eV$	½
= 25.59 eV	1

17. *Assertion (A) :* A proton and an alpha particle having same kinetic energy are moving in circular paths in a uniform magnetic field. The radii of their circular paths will be equal.

Reason (R) : The centripetal force required to move a charged particle in a circle does not depend on the magnitude of the magnetic field.

17. *Assertion (A) :* A proton and an alpha particle having same kinetic energy are moving in circular paths in a uniform magnetic field. The radii of their circular paths will be equal.

Reason (R) : The centripetal force required to move a charged particle in a circle does not depend on the magnitude of the magnetic field.

(c) Assertion (A) is true, but Reason (R) is false.

2. An electron with velocity $\vec{v} = (v_x \hat{i} + v_y \hat{j})$ moves through a magnetic field $\vec{B} = (B_x \hat{i} - B_y \hat{j})$. The force $\vec{F}$ on the electron is : (e is the magnitude of its charge)

(a) $-e(v_x B_y - v_y B_x)\hat{k}$

(b) $e(v_x B_y - v_y B_x)\hat{k}$

(c) $-e(v_x B_y + v_y B_x)\hat{k}$

(d) $e(v_x B_y + v_y B_x)\hat{k}$

2. An electron with velocity $\vec{v} = (v_x \hat{i} + v_y \hat{j})$ moves through a magnetic field $\vec{B} = (B_x \hat{i} - B_y \hat{j})$. The force $\vec{F}$ on the electron is : (e is the magnitude of its charge)

(a) $-e(v_x B_y - v_y B_x)\hat{k}$ (b) $e(v_x B_y - v_y B_x)\hat{k}$

(c) $-e(v_x B_y + v_y B_x)\hat{k}$ (d) $e(v_x B_y + v_y B_x)\hat{k}$

(d) $e(v_x B_y + v_y B_x)k$

16. **Assertion (A)** : The energy of a charged particle moving in a magnetic field does not change.

Reason (R) : It is because the work done by the magnetic force on the charge moving in a magnetic field is zero.

16. **Assertion (A)** : The energy of a charged particle moving in a magnetic field does not change.

Reason (R) : It is because the work done by the magnetic force on the charge moving in a magnetic field is zero.

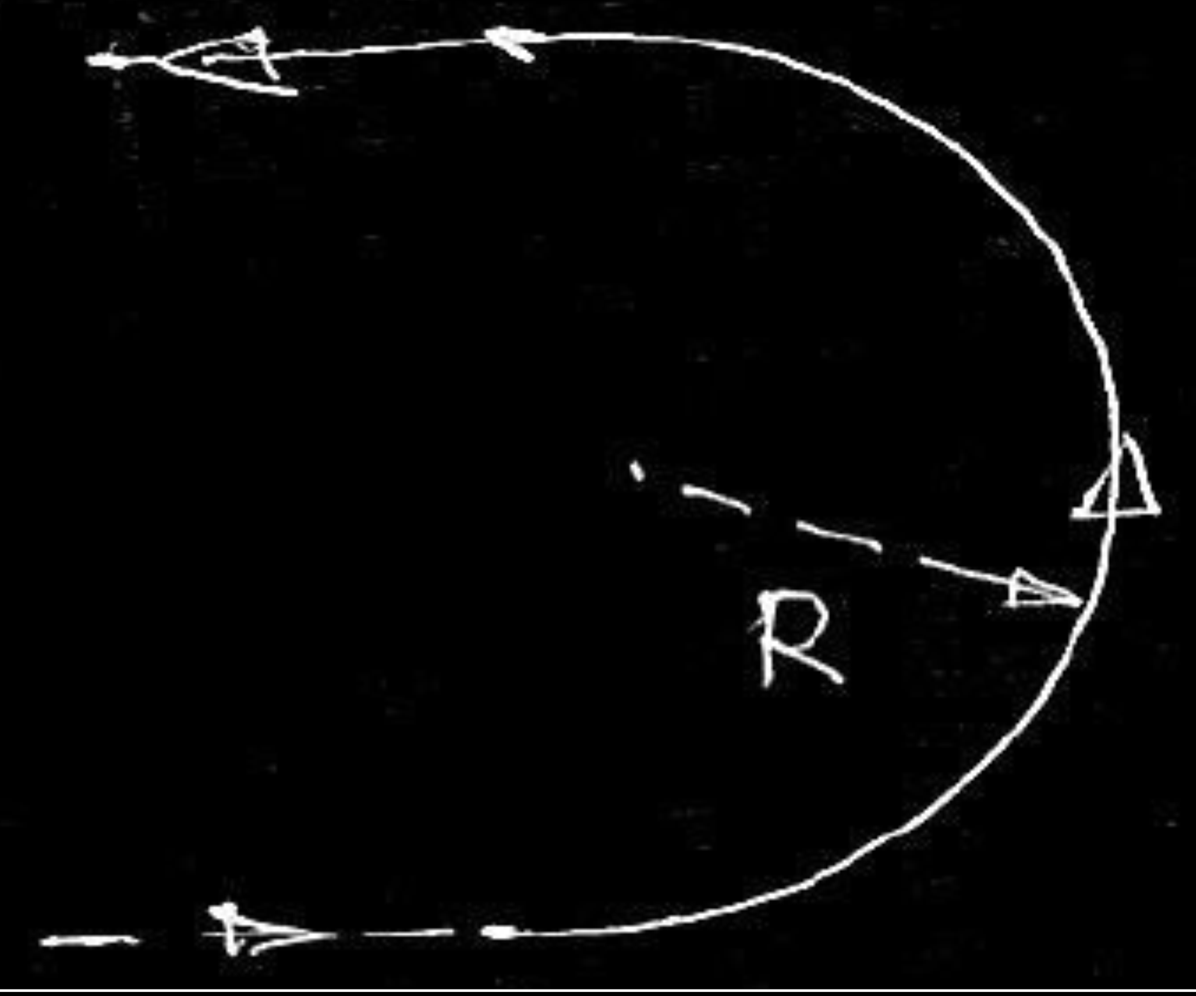
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).

27. An electron is moving with a velocity $\vec{v} = \left(3 \times 10^6 \frac{m}{s}\right)\hat{i}$. It enters a region of magnetic field $\vec{B} = (91 \text{ mT})\hat{k}$.

(a) Calculate the magnetic force $\vec{F}_B$ acting on electron and the radius of its path.

(b) Trace the path described by it.

(a) Calculation for magnetic force and radius	2
(b) Tracing the path	1

(a) $\vec{F}_B = q(\vec{\mathrm{v}} \times \vec{\mathrm{B}})$ $= -\ 1.6\times10^{-19}\left[(3\times10^{6}\,\hat{i})\times(91\times10^{-3}\,\hat{k})\right]$	½
$=1.6\times10^{-19}\left[3\times10^{6}\times91\times10^{-3}\right]\hat{j}$ $=4.368\times10^{-14}\,\hat{j}\ \ \mathrm{N}$	½
$r=\dfrac{m\mathrm{v}}{qB}$	½
$r=\dfrac{9.1\times10^{-31}\times3\times10^{6}}{1.6\times10^{-19}\times91\times10^{-3}}\ \mathrm{m}$ $r=1.875\times10^{-4}\ \mathrm{m}$	½
(b) Anticlockwise circular path  	1

2. A particle of mass m and charge q describes a circular path of radius R in a magnetic field. If its mass and charge were 2 m and $\frac{q}{2}$ respectively, the radius of its path would be

(A) $\frac{R}{4}$

(B) $\frac{R}{2}$

(C) 2 R

(D) 4 R

2. A particle of mass m and charge q describes a circular path of radius R in a magnetic field. If its mass and charge were 2 m and $\frac{q}{2}$ respectively, the radius of its path would be

(A) $\frac{R}{4}$

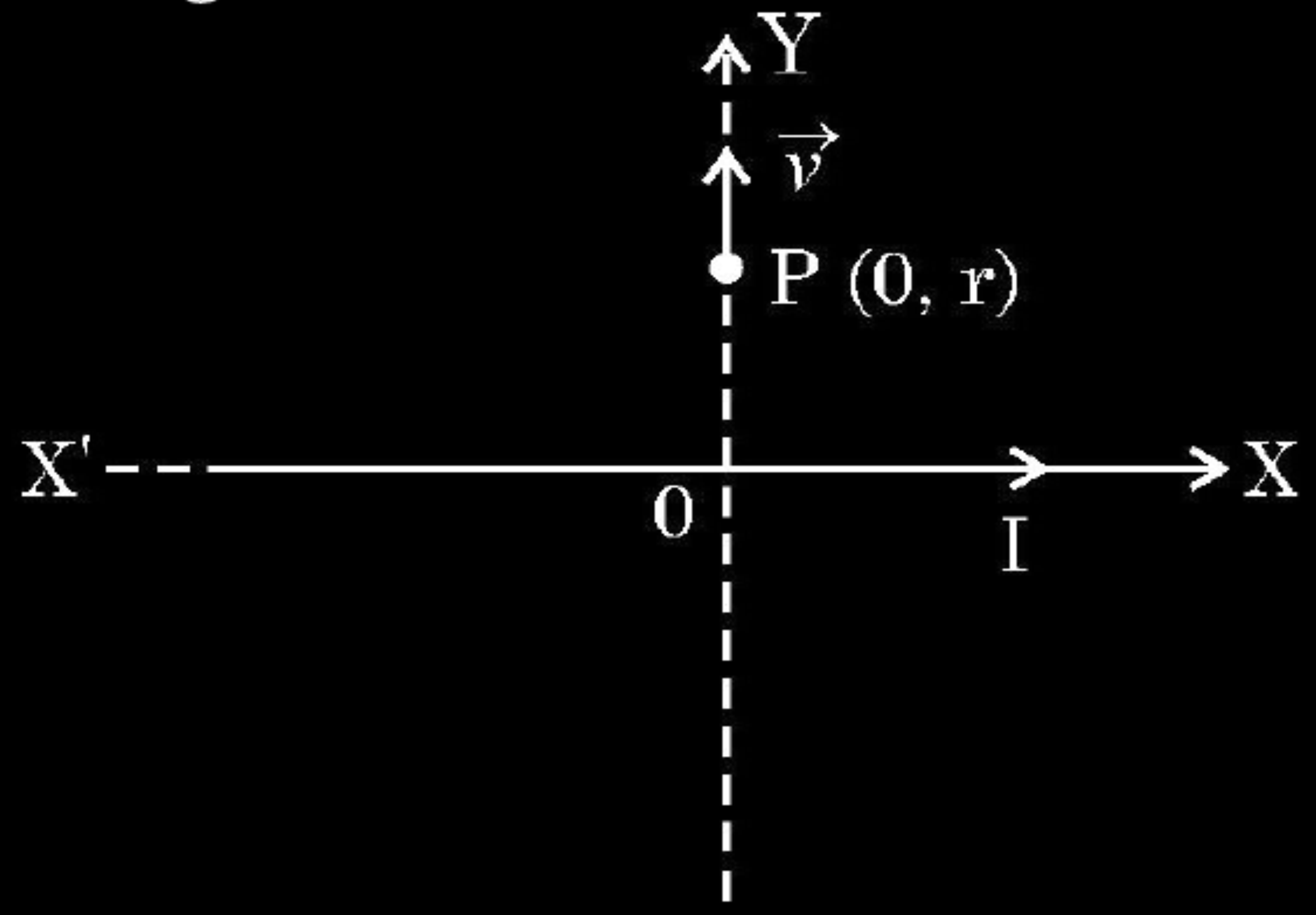
(B) $\frac{R}{2}$

(C) 2 R

(D) 4 R

(D) 4R

26. An infinite straight conductor is kept along $X'X$ axis and carries a current I. A charge q at point P(0, r) starts moving with velocity $\vec{v} = v_0 \hat{j}$ as shown in figure. Find the direction and magnitude of force initially experienced by the charge. **3**



Finding magnitude of force	2 ½
Finding direction of force	½

Magnetic field at P due to infinite straight conductor carrying current	
$$\vec{B} = \frac{\mu_0 I}{2\pi r}\hat{k}$$	½
Force on charge q in this magnetic field	
$$\vec{F} = q(\vec{v}\times\vec{B})$$	½
$$\vec{F} = q\left[(v_0\,\hat{j})\times\left(\frac{\mu_0 I}{2\pi r}\right)\hat{k}\right]$$	½
$$\vec{F} = \frac{\mu_0 q v_0 I}{2\pi r}\hat{i}$$	½
The magnitude of force $F = \frac{\mu_0 q v_0 I}{2\pi r}$	½
The direction of force on charge is along +ve X-axis.	½

5. A particle of mass m and charge q is moving with velocity $\vec{v} = v_x \hat{i} + v_y \hat{j}$.

If it is subjected to a magnetic field $\vec{B} = B_0 \hat{i}$, it will move in a – **1**

(A) straight line path

(B) circular path

(C) helical path

(D) parabolic path

5. A particle of mass m and charge q is moving with velocity $\vec{v} = v_x \hat{i} + v_y \hat{j}$.

If it is subjected to a magnetic field $\vec{B} = B_0 \hat{i}$, it will move in a – **1**

(A) straight line path (B) circular path

(C) helical path (D) parabolic path

(C) Helical path.

15. *Assertion (A) :* A proton and an electron enter a uniform magnetic field $\vec{B}$ with the same momentum $\vec{p}$ such that $\vec{p}$ is perpendicular to $\vec{B}$. They describe circular paths of the same radius.

Reason (R) : In a magnetic field, orbital radius r is equal to $\frac{p}{qB}$.

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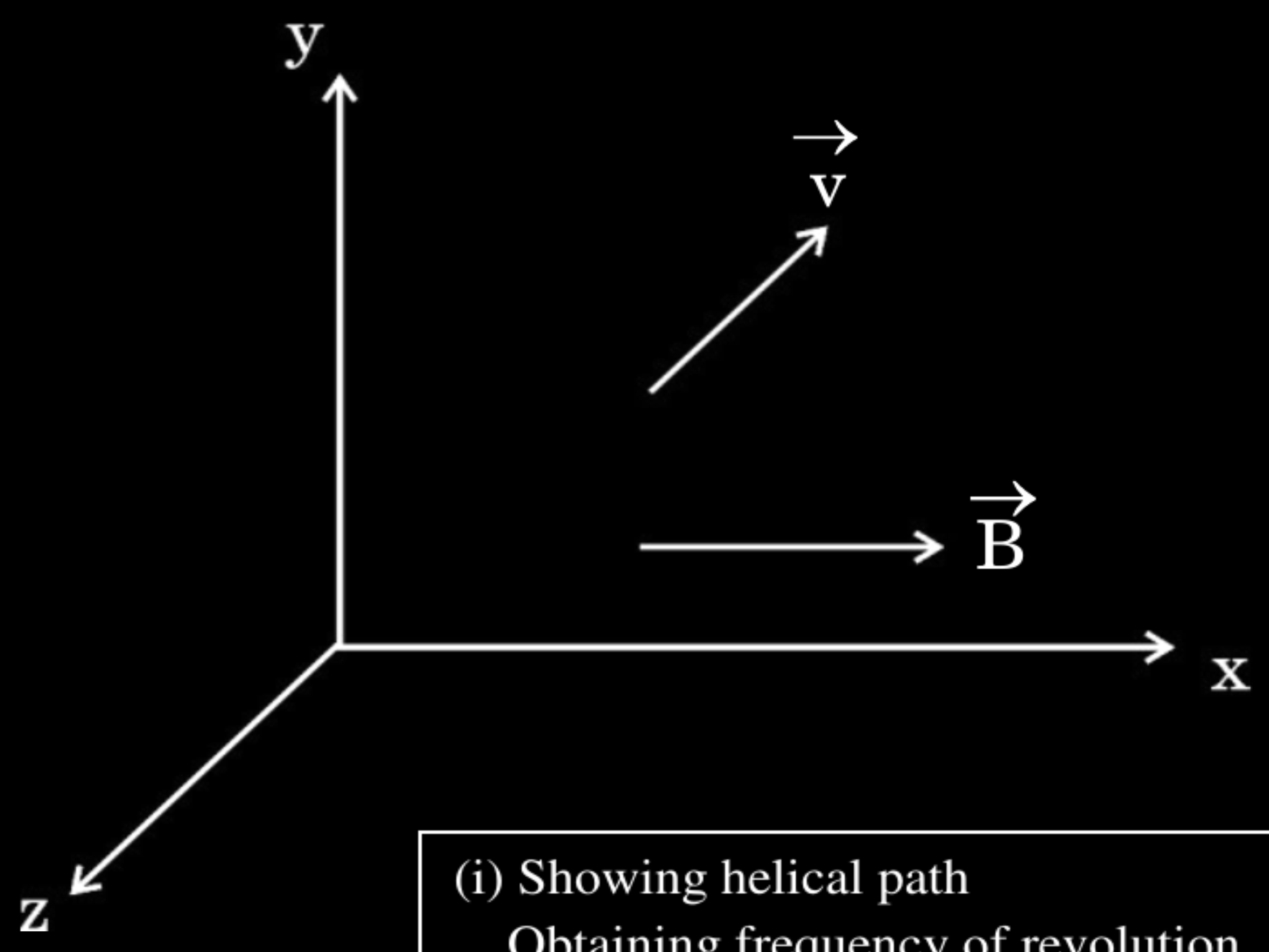
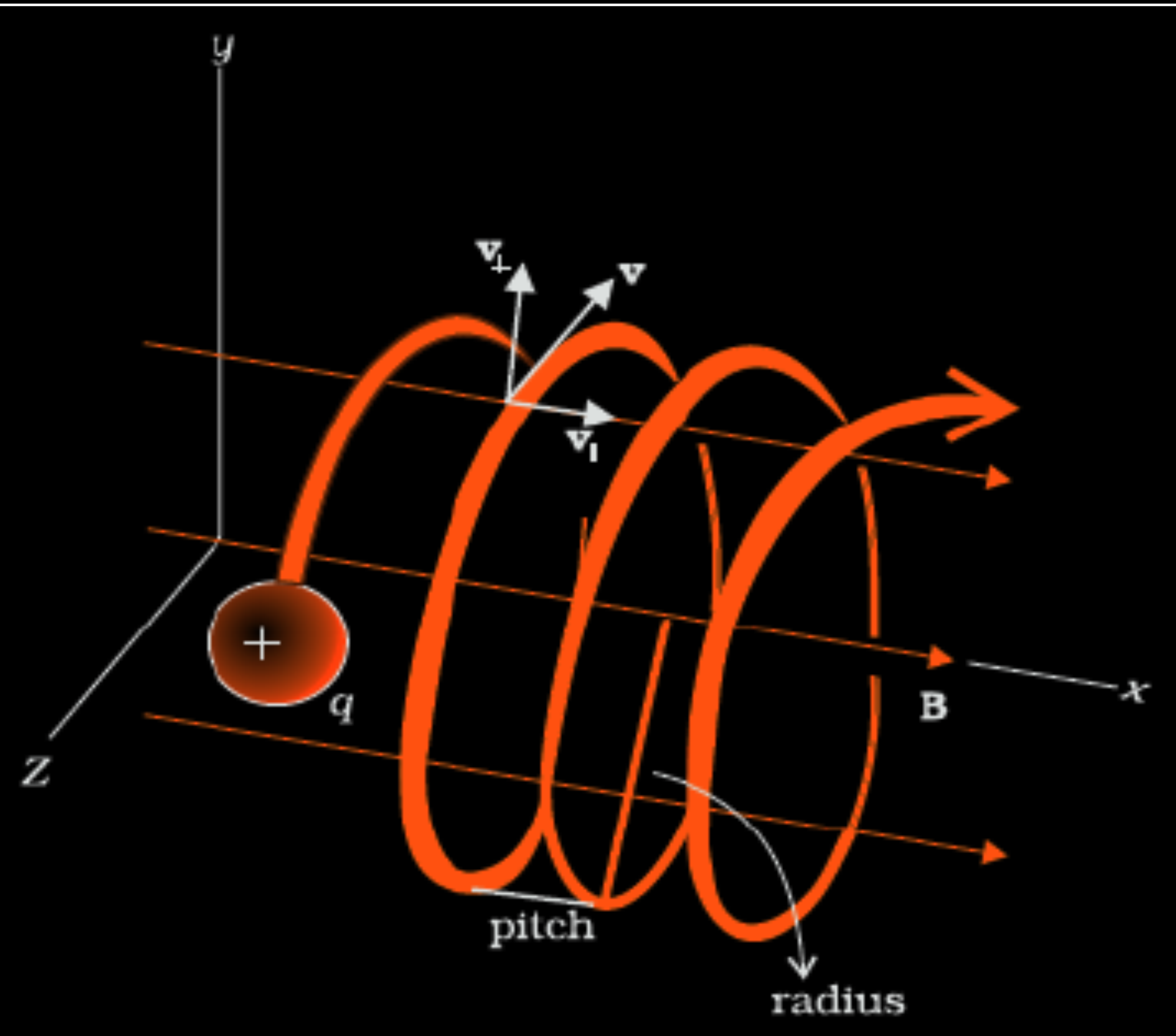
Reason (R) : In a magnetic field, orbital radius r is equal to $\frac{p}{qB}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion(A)

33. (a) (i) A particle of mass m and charge q is moving with a velocity $\vec{v}$ in a magnetic field $\vec{B}$ as shown in the figure. Show that it follows a helical path. Hence, obtain its frequency of revolution.



(i) Showing helical path	1 ½
Obtaining frequency of revolution	1 ½



½

$v_\perp = v\sin\theta$ is perpendicular to $\vec{B}$ and

$v_\parallel = v\cos\theta$ is parallel to $\vec{B}$

Due to $v_\perp$ the charge describes circular path and $v_\parallel$ pushes it in the direction of $\vec{B}$. Therefore under the combined effect of two components the charged particle describes helical path, as shown in the figure. **1**

The centripetal force

$$\frac{mv_\perp^{\ 2}}{r} = B\,qv_\perp$$ ½

$$v_\perp = \frac{Bqr}{m} \qquad (v_\perp = v\sin\theta)$$ ½

Time period = $T = \dfrac{2\pi r}{v_\perp}$

$$= \frac{2\pi m}{Bq}$$

$$frequency\, v = \frac{1}{T} = \frac{Bq}{2\pi m}$$ ½

14. *Assertion (A) :* An electron and a proton enter with the same momentum $\vec{p}$ in a magnetic field $\vec{B}$ such that $\vec{p} \perp \vec{B}$. Then both describe a circular path of the same radius.

Reason (R) : The radius of the circular path described by the charged particle (charge q, mass m) moving in the magnetic field $\vec{B}$ is given by $r = \dfrac{mv}{qB}$.

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(A) Both Assertion (A) and Reason (R) are true and Reason(R) is the correct explanation of the Assertion (A)

4. Two charged particles, P and Q, each having charge q but of masses m_1 and m_2, are accelerated through the same potential difference V. They enter a region of magnetic field $\vec{B}\,(\perp \vec{v})$ and describe the circular paths of radii a and b respectively. Then $\left(\frac{m_1}{m_2}\right)$ is equal to :

(A) $\frac{a}{b}$ (B) $\frac{b}{a}$ (C) $\left(\frac{a}{b}\right)^2$ (D) $\left(\frac{b}{a}\right)^2$

4. Two charged particles, P and Q, each having charge q but of masses m_1 and m_2, are accelerated through the same potential difference V. They enter a region of magnetic field $\vec{B}(\perp \vec{v})$ and describe the circular paths of radii a and b respectively. Then $\left(\frac{m_1}{m_2}\right)$ is equal to :

(A) $\frac{a}{b}$ (B) $\frac{b}{a}$ (C) $\left(\frac{a}{b}\right)^2$ (D) $\left(\frac{b}{a}\right)^2$

(C) $\left(\frac{a}{b}\right)^2$

18. An electron is passing through a region and experiences no force. Under what condition is it possible when the region has (a) only the magnetic field, and (b) both the electric and the magnetic fields ? Justify your answers. *2*

Condition for experiencing no force on electron in	
(a) Only magnetic field and justification	½ +½
(b) In electric and magnetic field both and justification	½ +½

18. An electron is passing through a region and experiences no force. Under what condition is it possible when the region has (a) only the magnetic field, and (b) both the electric and the magnetic fields ? Justify your answers. *2*

(a) Force on an electron in magnetic field is zero when it moves parallel/ antiparallel to the direction of magnetic field.	½
$F = qvB \sin\theta$	
Hence $F = 0$, when $\theta = 0^0, \theta = 180^0$	½
(b) Force is zero when electron moves in a crossed field of electric and magnetic fields with condition.	½
$\vec{F}_E + \vec{F}_B = 0$	½
$v = \dfrac{E}{B}$	

Condition for experiencing no force on electron in	
(a) Only magnetic field and justification	½ +½
(b) In electric and magnetic field both and justification	½ +½

22. A particle of mass m and charge q is moving in a magnetic field $\vec{B}$ with a velocity $\vec{v}$. Discuss, giving reasons, the shape of its trajectory when the angle between $\vec{v}$ and $\vec{B}$ is : *3*

(a) $0°$

(b) $90°$

(c) $120°$

Shape of trajectory with reasons when the angle between $\vec{V}$ & $\vec{B}$ is	
(a) 0^0	½ + ½
(b) 90^0	½ + ½
(c) 120^0	½ + ½

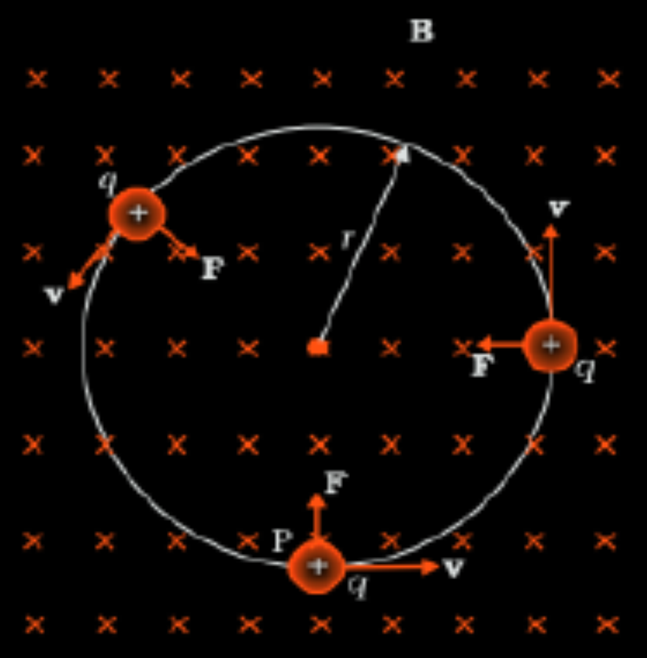
(a) We know $\vec{F} = q(\vec{v} \times \vec{B})$

$\vec{v} \longrightarrow \quad \vec{B} \longrightarrow$

Alternatively:- The particle moves in a straight line. ½

As the angle between $\vec{v}$ & $\vec{B}$ is 0^0 hence $\vec{F} = 0$ ½

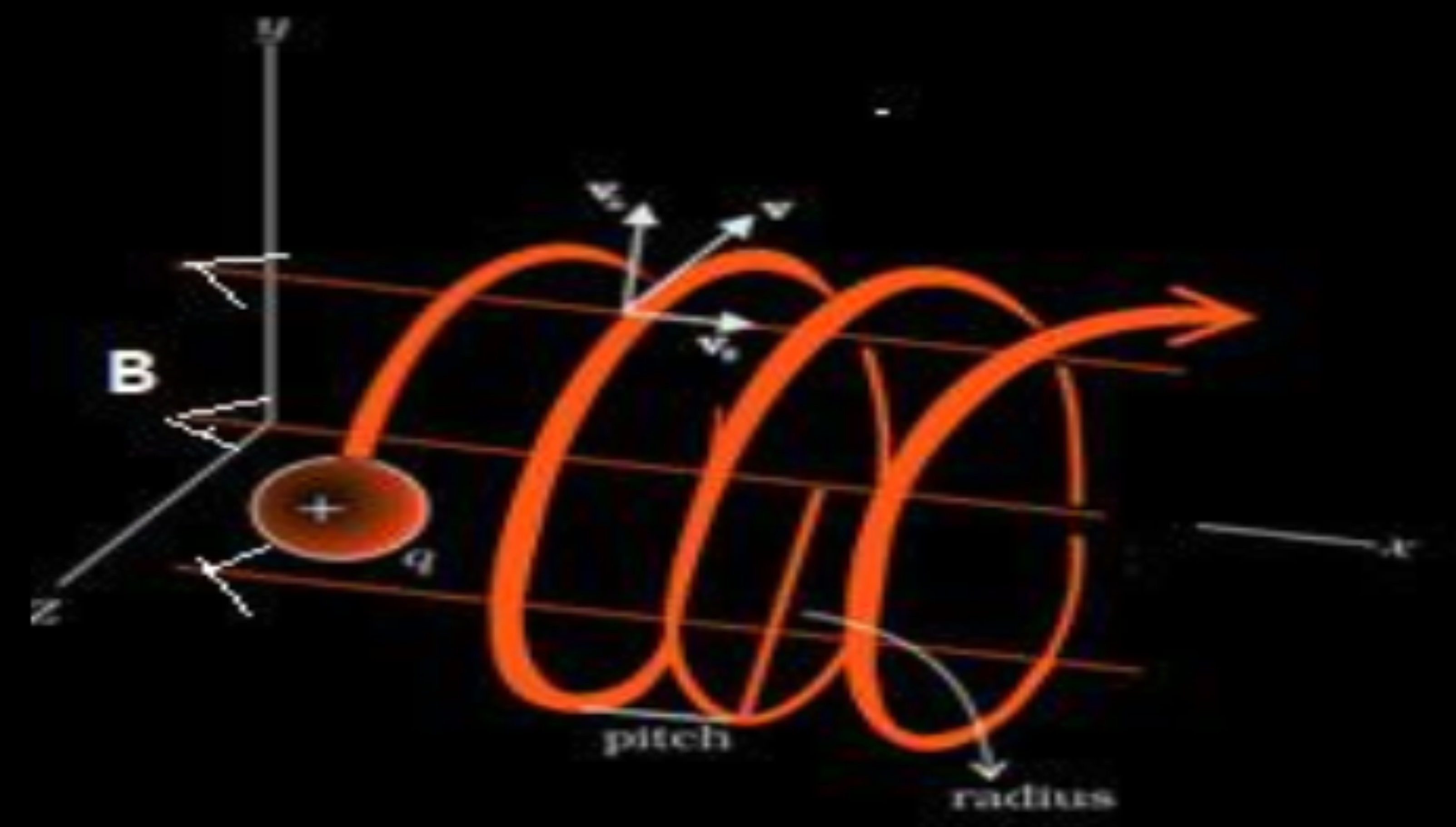
(b)



Alternatively- The particle moves in a circular path. ½

F is perpendicular to both $\vec{v}$ & $\vec{B}$ ½

(c)



Alternatively The particle follows a helical path. ½

$F = q\,v\,B\,Sin\theta$

Component of velocity parallel to magnetic field tends to move the particle along linear path while the component perpendicular to magnetic field tends to move the particle in circular path. As a consequence the particle moves in a helical path. ½

15. *Assertion (A) :* A magnetic field does not interact with a moving charge.

Reason (R) : A moving charge does not produce a magnetic field.

15. *Assertion (A) :* A magnetic field does not interact with a moving charge.

Reason (R) : A moving charge does not produce a magnetic field.

(D) Assertion (A) is false and Reason (R) is also false.

32. (a) (i) Discuss the motion of a charged particle of mass m and charge q in a uniform magnetic field $\vec{B}$ with initial velocity $\vec{v}$

(1) at an angle θ with field direction,

(2) perpendicular to the field direction (θ = 90°), and

(3) parallel to the field direction (θ = 0°).

(ii) An electron is moving along a circular path of radius 18.2×10^{-2} m in a uniform magnetic field 2×10^{-3} T. Calculate : *5*

(1) the speed of the electron, and

(2) the potential difference through which the electron must be accelerated to acquire this speed.

(i)Discussing the motion of charged particle when it is moving	
(1) at an angle θ with field direction	1
(2) Perpendicular to the field direction	1
(3) Parallel to the field direction	1
(ii) Calculating	
(1) The speed of electron	1
(2) Potential difference	1

(i) The force on a charge particle of mass m and charge q moving in a uniform magnetic field $\vec{B}$ velocity $\vec{v}$ $\vec{F} = q(\vec{v} \times \vec{B})$ (1) The particle follows a helical path. As the component of velocity parallel to magnetic field tends to move the particle along linear path while the component perpendicular to magnetic field tends to move the particle in circular path. Hence, the particle follows a helical path.	1
(2) The particle follows a circular path. As total magnetic force $(F = qvB)$	1
(3) The particle moves in a straight line. As force on particle will be zero.	1
(ii) (1) Speed of the electron $\mathrm{v} = \frac{\mathrm{qBr}}{\mathrm{m}}$	½
$\mathrm{v} = \frac{1.6\times10^{-19}\times2\times10^{-3}\times18.2\times10^{-2}}{9.1\times10^{-31}}$ $\mathrm{v} = 6.4\times10^{7}$ m/s	½
(2) Accelerating Potential Difference $V = \frac{m\mathrm{v}^2}{2e}$	½
$V = \frac{9.1\times10^{-31}\times(6.4\times10^{7})^2}{2\times1.6\times10^{-19}}$ =11.65 kV	½

14. *Assertion (A) :* A charged particle is moving with velocity v in x-y plane, making an angle $\theta \, (0 < \theta < \frac{\pi}{2})$ with x-axis. If a uniform magnetic field $\vec{B}$ is applied in the region, along y-axis, the particle will move in a helical path with its axis parallel to x-axis.

Reason (R) : The direction of the magnetic force acting on a charged particle moving in a magnetic field is along the velocity of the particle.

14. *Assertion (A) :* A charged particle is moving with velocity v in x-y plane, making an angle $\theta \left(0 < \theta < \frac{\pi}{2}\right)$ with x-axis. If a uniform magnetic field $\vec{B}$ is applied in the region, along y-axis, the particle will move in a helical path with its axis parallel to x-axis.

Reason (R) : The direction of the magnetic force acting on a charged particle moving in a magnetic field is along the velocity of the particle.

(D) Both Assertion (A) and Reason (R) are false.

7. A proton and an α-particle enter with the same velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$ such that $\vec{v} \perp \vec{B}$. The ratio of the radii of their paths is :

(A) 2

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) 4

7. A proton and an α-particle enter with the same velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$ such that $\vec{v} \perp \vec{B}$. The ratio of the radii of their paths is :

(A) 2

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) 4

(B) $\frac{1}{2}$

3. A particle of mass m and charge q moving with a velocity $\acute{V}$ at right angle to a magnetic field $\vec{B}$, experiences a force $\vec{F}$. Another particle of mass 2 m and charge 2q enters the same field $\vec{B}$ with a velocity $\frac{\vec{V}}{2}$, in the same direction. This particle will experience a force 1

(A) $4\vec{F}$

(B) $2\vec{F}$

(C) $\vec{F}$

(D) $\frac{\vec{F}}{2}$

3. A particle of mass m and charge q moving with a velocity $\acute{V}$ at right angle to a magnetic field $\vec{B}$, experiences a force $\vec{F}$. Another particle of mass 2 m and charge 2q enters the same field $\vec{B}$ with a velocity $\frac{\vec{V}}{2}$, in the same direction. This particle will experience a force **1**

(A) $4\vec{F}$

(B) $2\vec{F}$

(C) $\vec{F}$

(D) $\frac{\vec{F}}{2}$

(C) F

(b) (A) A particle of mass m and charge q is moving with velocity v in a circular path under the influence of a uniform magnetic field $\vec{B}$. Explain how will the path followed be affected when

(i) the strength of magnetic field is decreased.

(ii) the velocity of the particle is decreased.

(iii) the velocity of charged particle is such that one of its component is perpendicular to B and another component is parallel to $\vec{B}$.

(iv) an electric field is applied in such a way that electrostatic force balances the magnetic force on the charged particle.

(v) the magnetic field is suddenly removed.

In each of the above cases, other factors remain the same.

(A) Explaining, the reason for changing the path of a charge particle, when	
(i) the strength of magnetic field is decreased	½
(ii) the velocity of charge particle is decreased	½
(iii) the component of velocity of charged particle is along magnetic field and perpendicular to magnetic field	½
(iv) electrostatic force balance, the magnetic force	½
(v) the magnetic field is switched off	½
(B) Deriving the expression for torque	2 ½

(i) Radius of circular path in magnetic field is $$r = \frac{mv}{qB}$$ $$r \propto \frac{1}{B}$$ If B decreases , r increases.	½
(ii) $r \propto$ v If v decreases , r decreases.	½
(iii) Under the influence of these two components the charge particle follow helical path.	½
(iv) The charge particle will move in the field undeviated.	½
(v) The charge particle will fly off tangently along the direction of velocity.	½

6. A charged particle gains a speed of 10^6 ms^{-1}, when accelerated from rest through a potential difference 10 kV. It enters a region of magnetic field of 0·4 T such that $\vec{v} \perp \vec{B}$. The radius of circular path described by it is :

(A) 2·5 cm (B) 5 cm

(C) 8 cm (D) 10 cm

6. A charged particle gains a speed of 10^6 ms^{-1}, when accelerated from rest through a potential difference 10 kV. It enters a region of magnetic field of 0·4 T such that $\vec{v} \perp \vec{B}$. The radius of circular path described by it is :

(A) 2·5 cm (B) 5 cm

(C) 8 cm (D) 10 cm

(B) 5 cm

3. A particle of mass m and charge q moves along y-axis in a region in which a uniform magnetic field $\vec{B}$ is pointing along x-axis. The Lorentz force acting on the charge will point along :

(A) x-axis (B) y-axis

(C) z-axis (D) negative z-axis

3. A particle of mass m and charge q moves along y-axis in a region in which a uniform magnetic field $\vec{B}$ is pointing along x-axis. The Lorentz force acting on the charge will point along :

(A) x-axis (B) y-axis

(C) z-axis (D) negative z-axis

(D) Negative Z axis

3. A particle with charge q moving with velocity $\vec{v} = v_0 \hat{i}$ enters a region with magnetic field $\vec{B} = B_1 \hat{j} + B_2 \hat{k}$. The magnitude of force experienced by the particle is :

(A) $qv_0 (B_1 + B_2)$

(B) $q\sqrt{v_0 (B_1 + B_2)}$

(C) $q v_0 \sqrt{(B_1^2 + B_2^2)}$

(D) $q \sqrt{v_0 (B_1^2 + B_2^2)}$

3. A particle with charge q moving with velocity $\vec{v} = v_0\hat{i}$ enters a region with magnetic field $\vec{B} = B_1\hat{j} + B_2\hat{k}$. The magnitude of force experienced by the particle is :

(A) $qv_0(B_1 + B_2)$

(B) $q\sqrt{v_0(B_1 + B_2)}$

(C) $qv_0\sqrt{(B_1^2 + B_2^2)}$

(D) $q\sqrt{v_0(B_1^2 + B_2^2)}$

(C) $qv_0\sqrt{B_1^2 + B_2^2}$

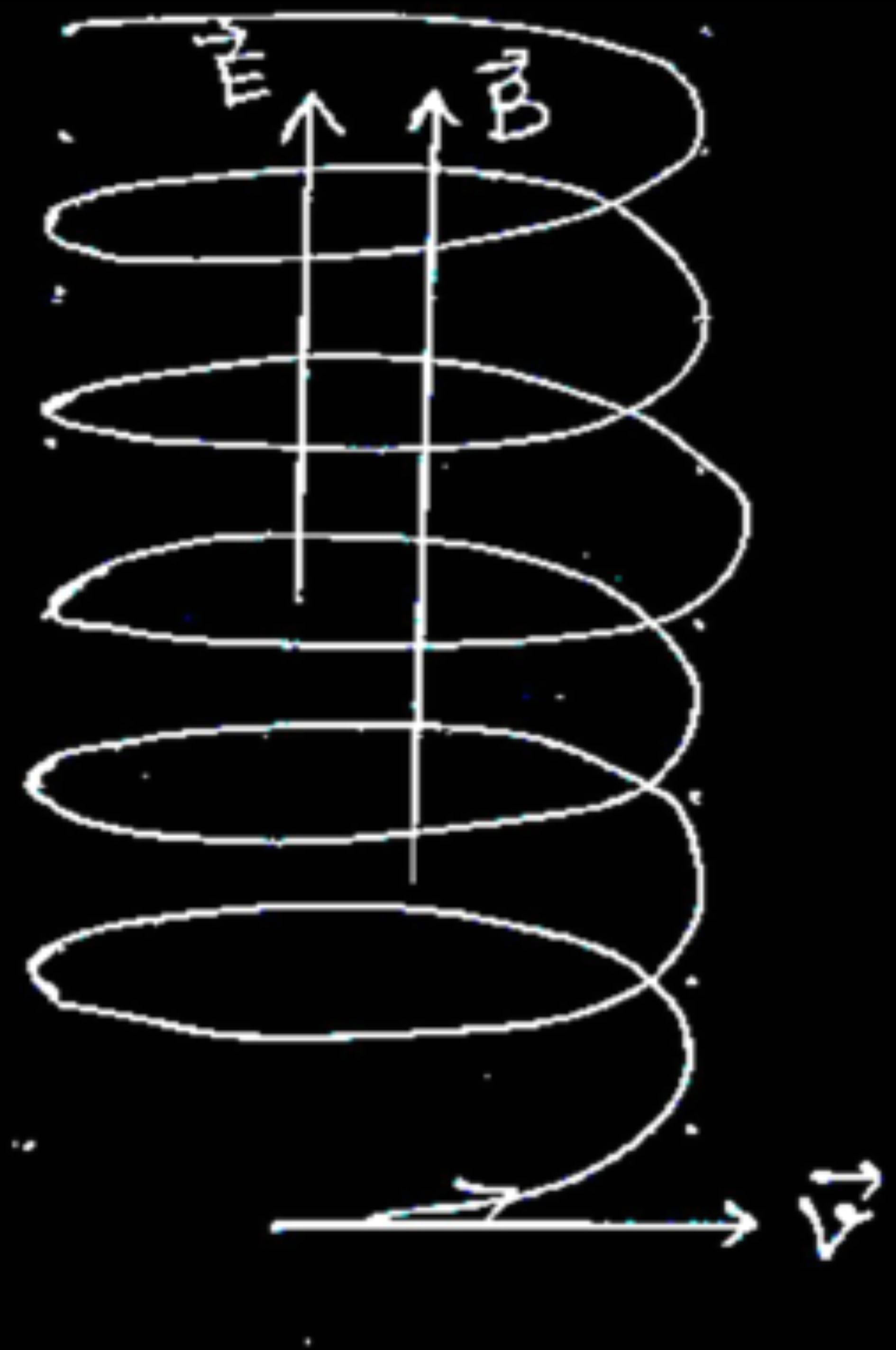
23. (a) A charged particle q moving with a velocity $\vec{v}$ is subjected to a uniform magnetic field $\vec{B}$ acting perpendicular to $\vec{v}$. If a uniform electric field $\vec{E}$ is also set up in the region along the direction of $\vec{B}$, describe the path followed by the particle and draw its shape.

(b) How will the magnetic field inside a long solenoid be affected when :

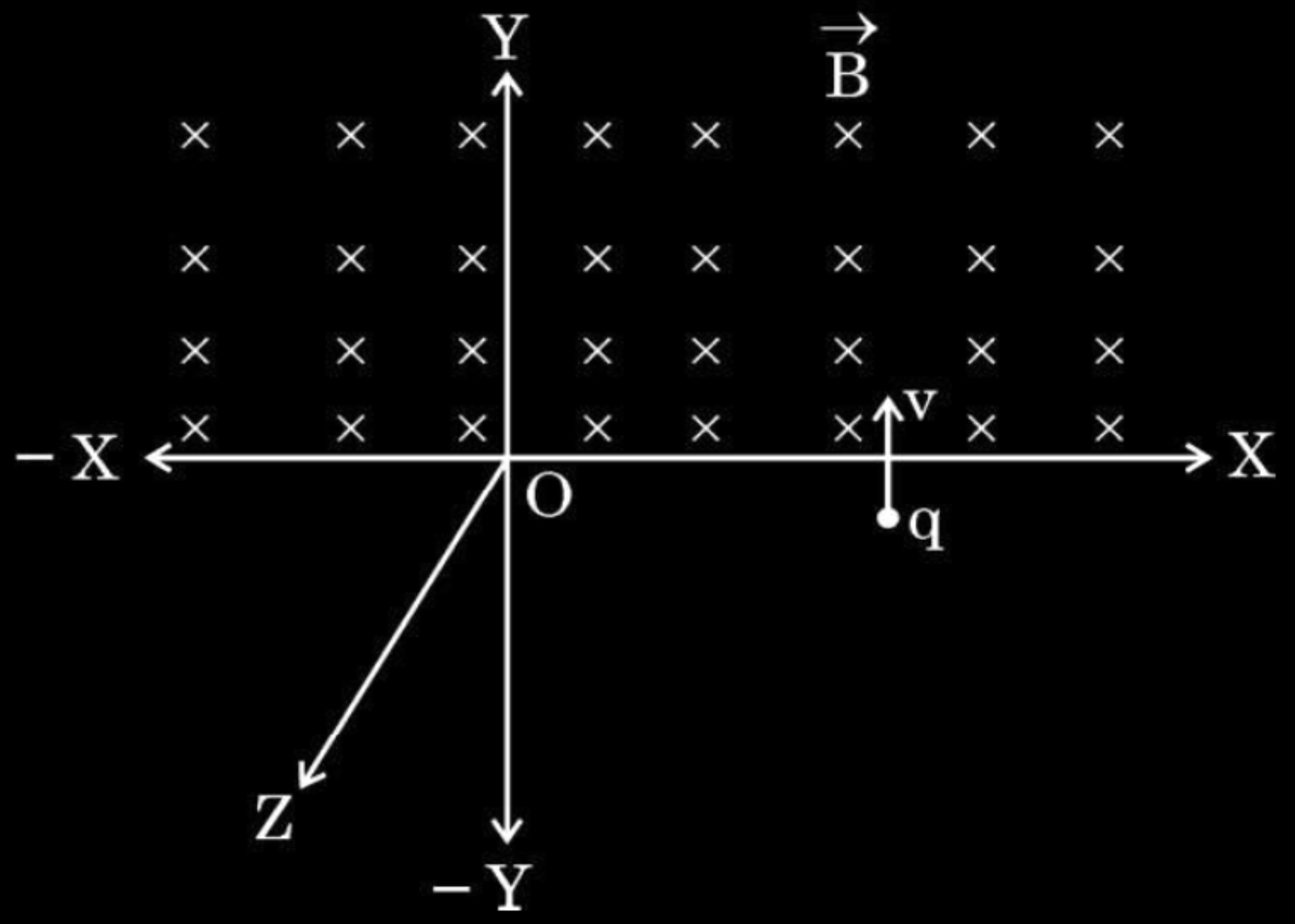
(i) the radius of the turns of the solenoid is increased,

(ii) the length of solenoid as well as the total number of its turns are doubled ? *3*

(a)	
• Explanation of a path followed by the particle	1
• Shape of path	1
(b) Effect on magnetic field when	
(i) Radius of turns of solenoid is increased	½
(ii) Length and number of turns are doubled	½

(a) Due to magnetic field particle will follow circular path and due to electric field, particle will accelerate along the electric field. As a result particle will follow a helical path with constant radius but increasing pitch.	1
	1
(b) (i) $B = \frac{\mu_o NI}{l}$ No change	½
(ii) $\frac{N}{l} = \frac{2N}{2l}$ No Change	½

3. A particle having charge +q enters a uniform magnetic field $\vec{B}$ as shown in the figure. The particle will describe :



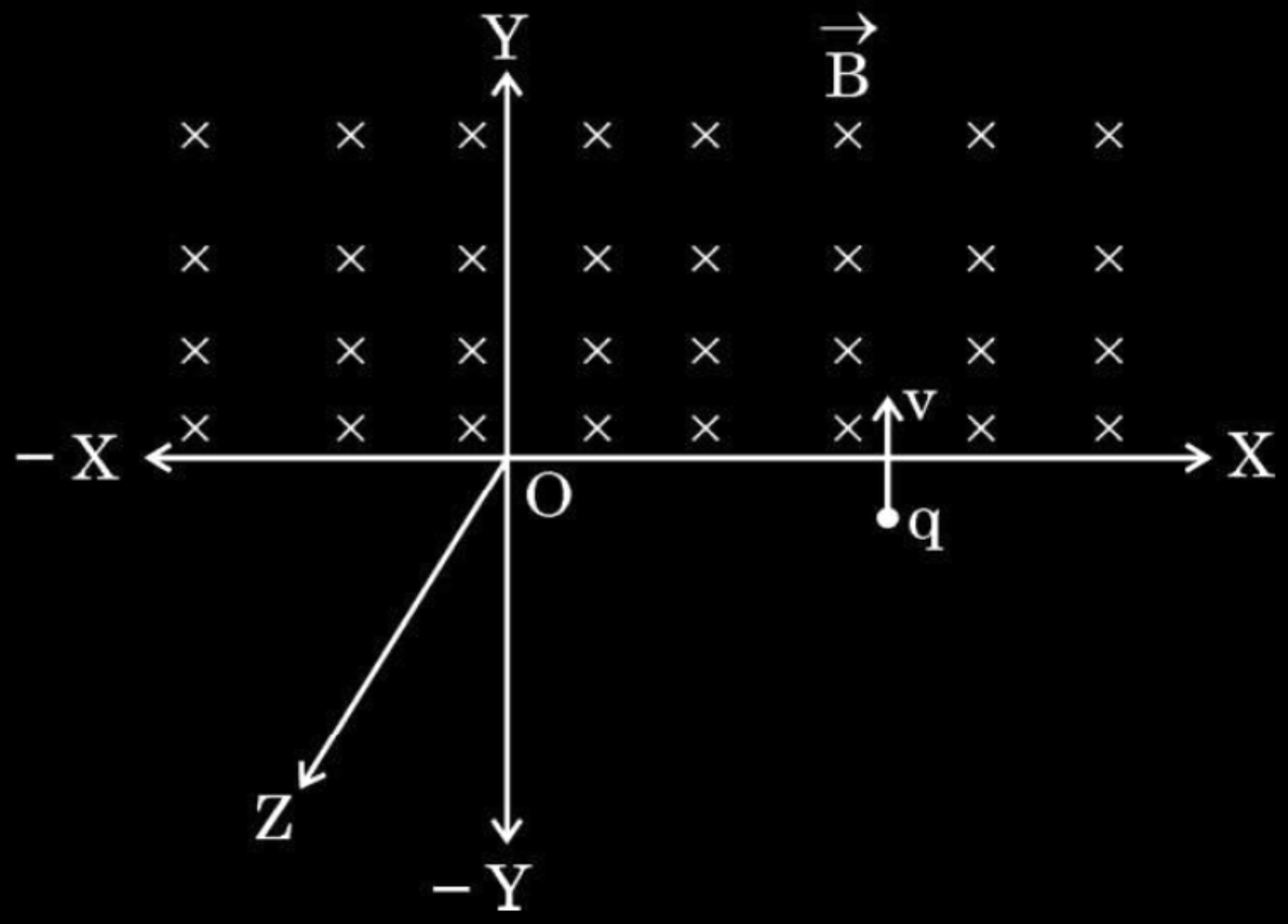
(A) a circular path in XZ plane

(B) a semicircular path in XY plane

(C) a helical path with its axis parallel to Y-axis

(D) a semicircular path in YZ plane

3. A particle having charge +q enters a uniform magnetic field $\vec{B}$ as shown in the figure. The particle will describe :



(A) a circular path in XZ plane

(B) a semicircular path in XY plane

(C) a helical path with its axis parallel to Y-axis

(D) a semicircular path in YZ plane

(B) a semicircular path in XY plane

27. (a) In a region of a uniform electric field $\vec{E}$, a negatively charged particle is moving with a constant velocity $\vec{v} = -v_o \hat{i}$ near a long straight conductor coinciding with XX' axis and carrying current I towards –X axis. The particle remains at a distance d from the conductor. **3**

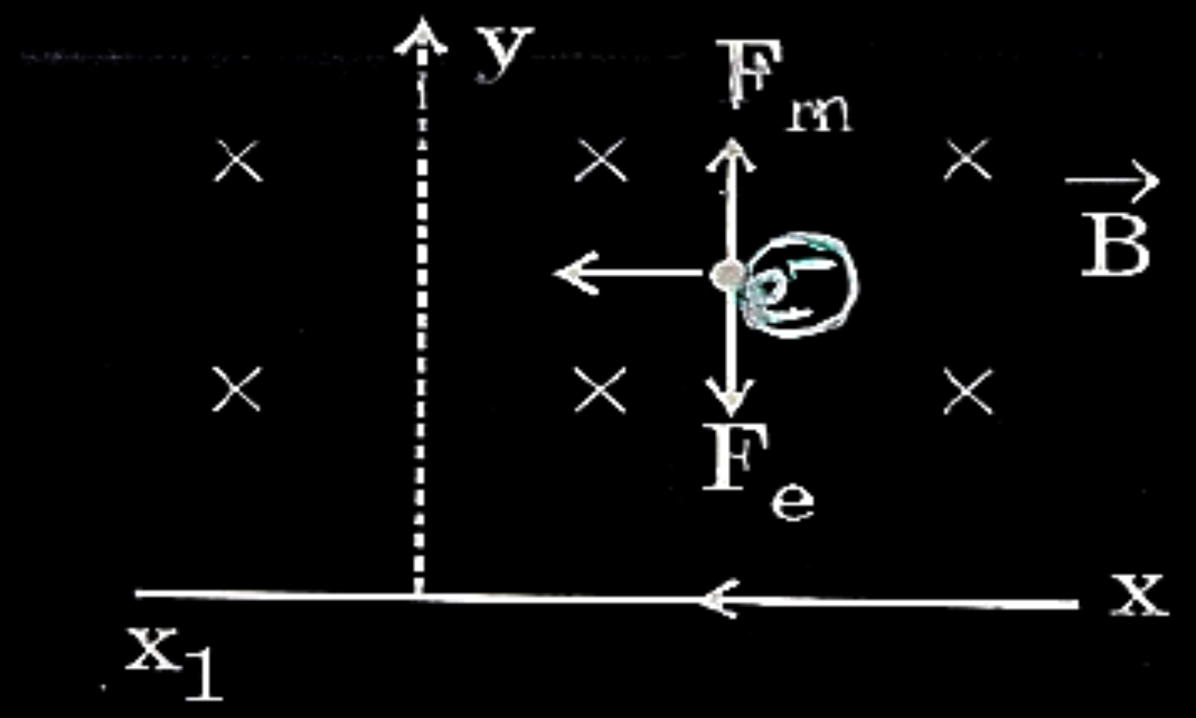
(i) Draw diagram showing direction of electric and magnetic fields.

(ii) What are the various forces acting on the charged particle ?

(iii) Find the value of v_o in terms of E, d and I.

(i) Diagram showing direction of electric and magnetic fields	1
(ii) Naming forces acting on the charged particle	1
(iii) Finding the value of υ_0	1

(i)



1

(ii) Electric force ½

Magnetic force ½

Alternatively: -

$F_E = eE$

$F_B = evB$

(iii) $ev_oB = eE$ ½

$$v_o \times \left[\frac{\mu_o I}{2\pi d}\right] = E$$

$$v_o = \frac{(2\pi d)E}{\mu_o I}$$ ½

31. (a) (i) What is the source of force acting on a current-carrying conductor placed in a magnetic field ? Obtain the expression for force acting between two long straight parallel conductors carrying steady currents and hence define 'ampere'. **5**

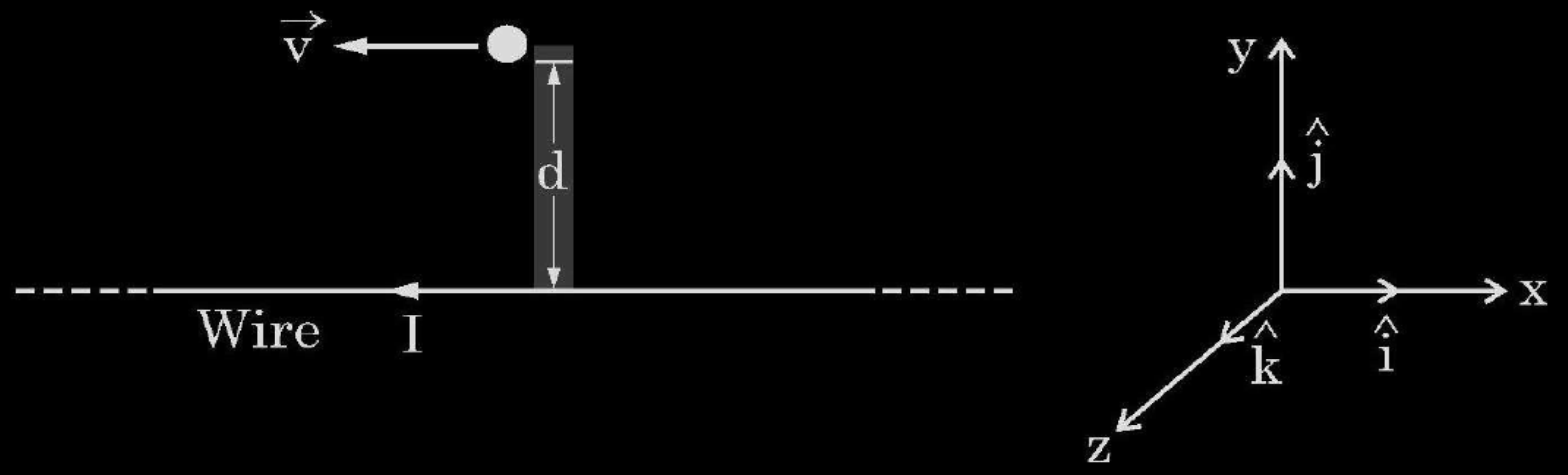
(ii) A point charge q is moving with velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$. Find the work done by the magnetic force on the charge.

(iii) Explain the necessary conditions in which the trajectory of a charged particle is helical in a uniform magnetic field.

(i) Source of force	½
Obtaining expression for force	1½
Definition of 'ampere'	1
(ii) Finding work done by the magnetic force	1
(iii) Necessary conditions	1

(i) The source of force is the interaction between the field produced by the current carrying conductor and the external field in which it is placed.	½
Two long parallel conductors a & b, separated by a distance d, carrying currents I_a and I_b, respectively. The magnetic field due to a, $B_a = \frac{\mu_0 I_a}{2\pi d}$ The force F_{ba}, is the force on a segment L of 'b' due to 'a'. $F_{ba} = I_b L B_a$ $= \frac{\mu_0 I_a I_b}{2\pi d} L$	½ 1
Definition – The 'ampere' is that value of steady current which, when maintained in each of the two very long, straight, parallel conductors of negligible cross-section, and placed one metre apart in vacuum, would produce on each of these conductors a force equal to $2\text{x}10^{-7}$ newton per metre of length. (ii) Work done by the magnetic force on the charge is zero as force is perpendicular to $\vec{v}$. (iii) The velocity ($\vec{v}$) is at an arbitrary angle θ w.r.t the magnetic field ($\vec{B}$).	1 1 1

23. A particle of charge q is moving with a velocity $\vec{v}$ at a distance 'd' from a long straight wire carrying a current 'I' as shown in figure. At this instant, it is subjected to a uniform electric field $\vec{E}$ such that the particle keeps moving undeviated. In terms of unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$, find – **3**



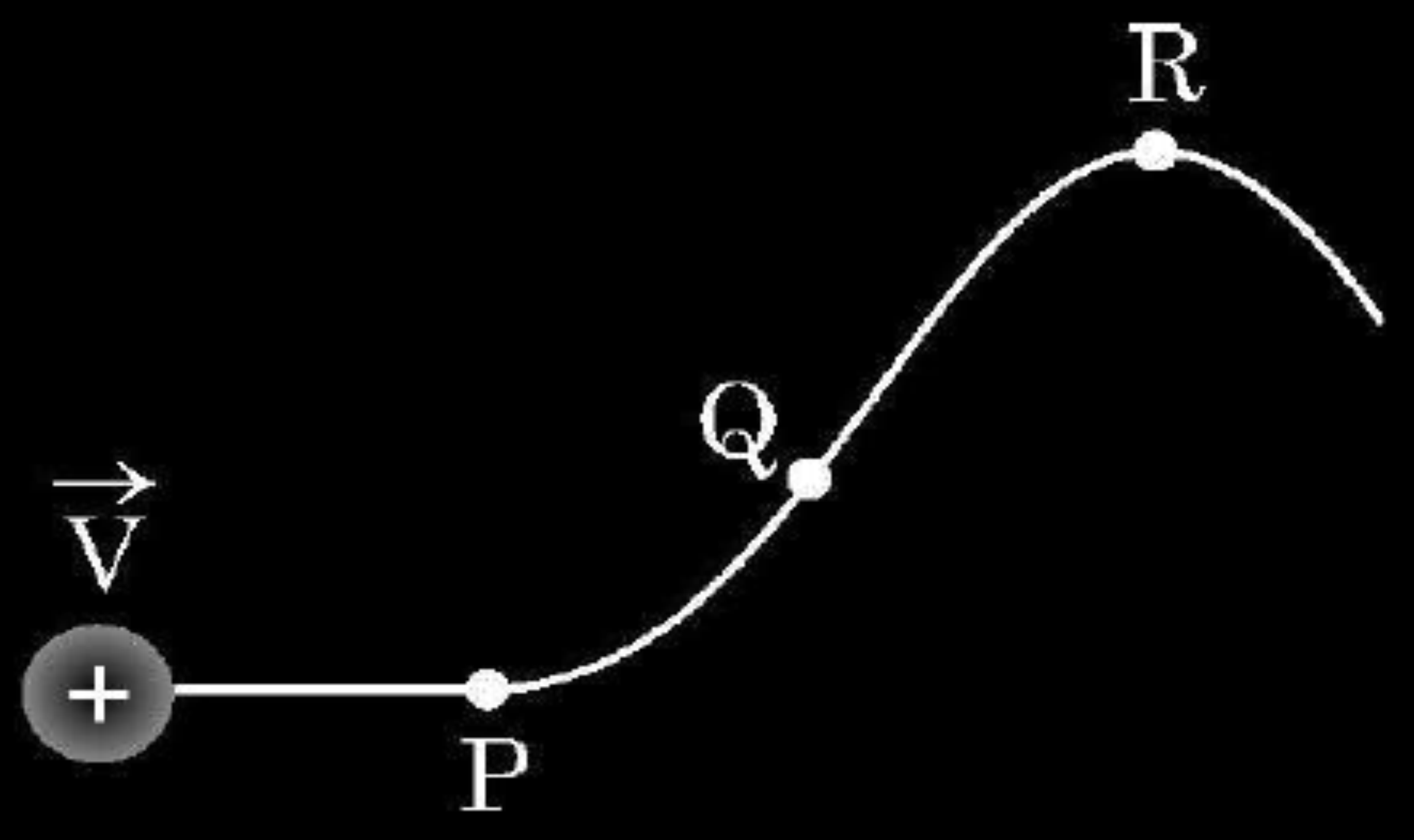
(a) the magnetic field $\vec{B}$,

(b) the magnetic force $\vec{F}_m$, and

(c) the electric field $\vec{E}$, acting on the charge.

Finding	
a) The magnetic field $\vec{B}$	1
b) The magnetic force $\vec{F}_m$	1
c) The electric field $\vec{E}$	1

33. (a) (i) A proton moving with velocity $\vec{V}$ in a non-uniform magnetic field traces a path as shown in the figure. **5**

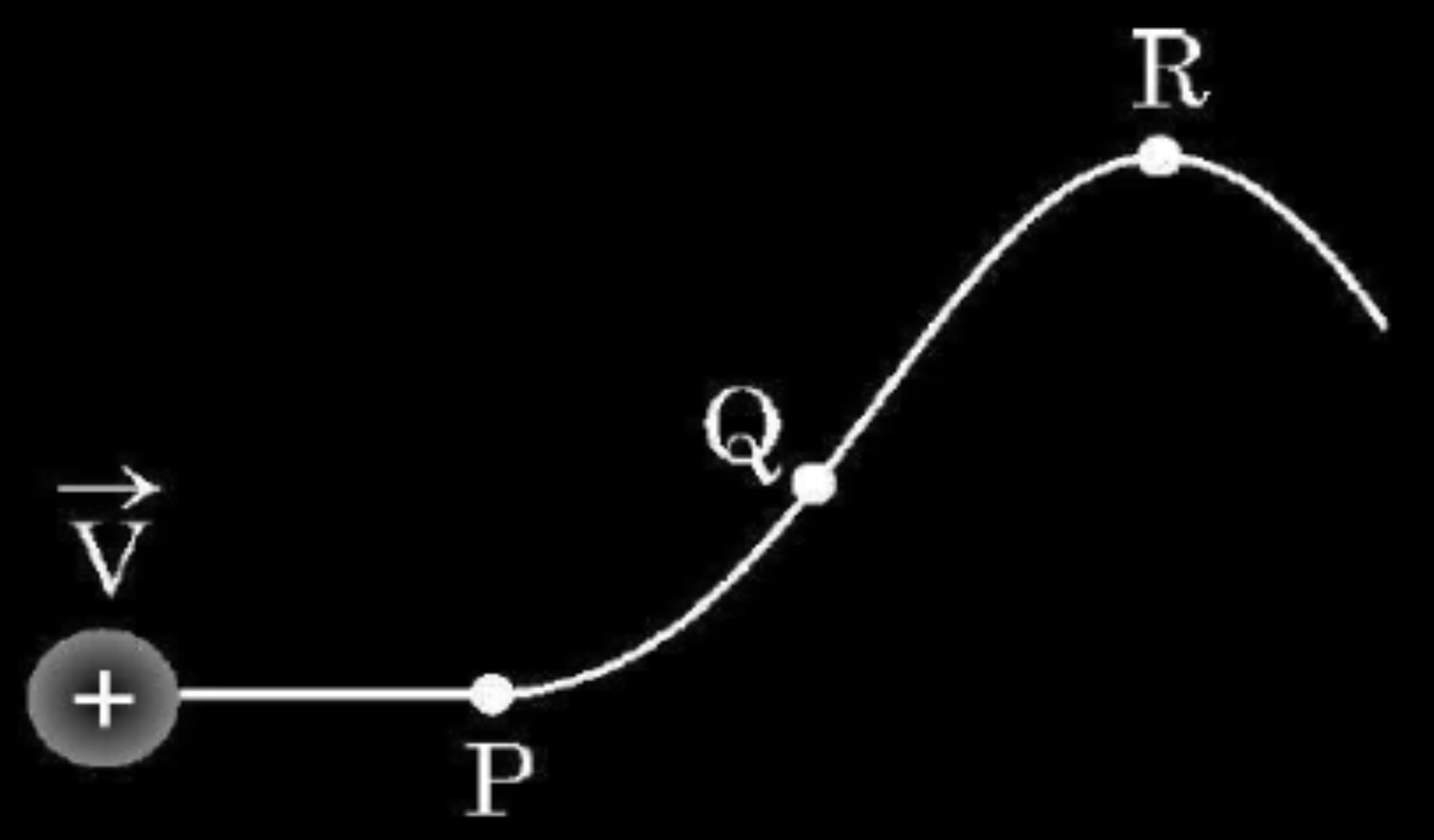


The path followed by the proton is always in the plane of the paper. What is the direction of the magnetic field in the region near points P, Q and R ? What can you say about relative magnitude of magnetic fields at these points ?

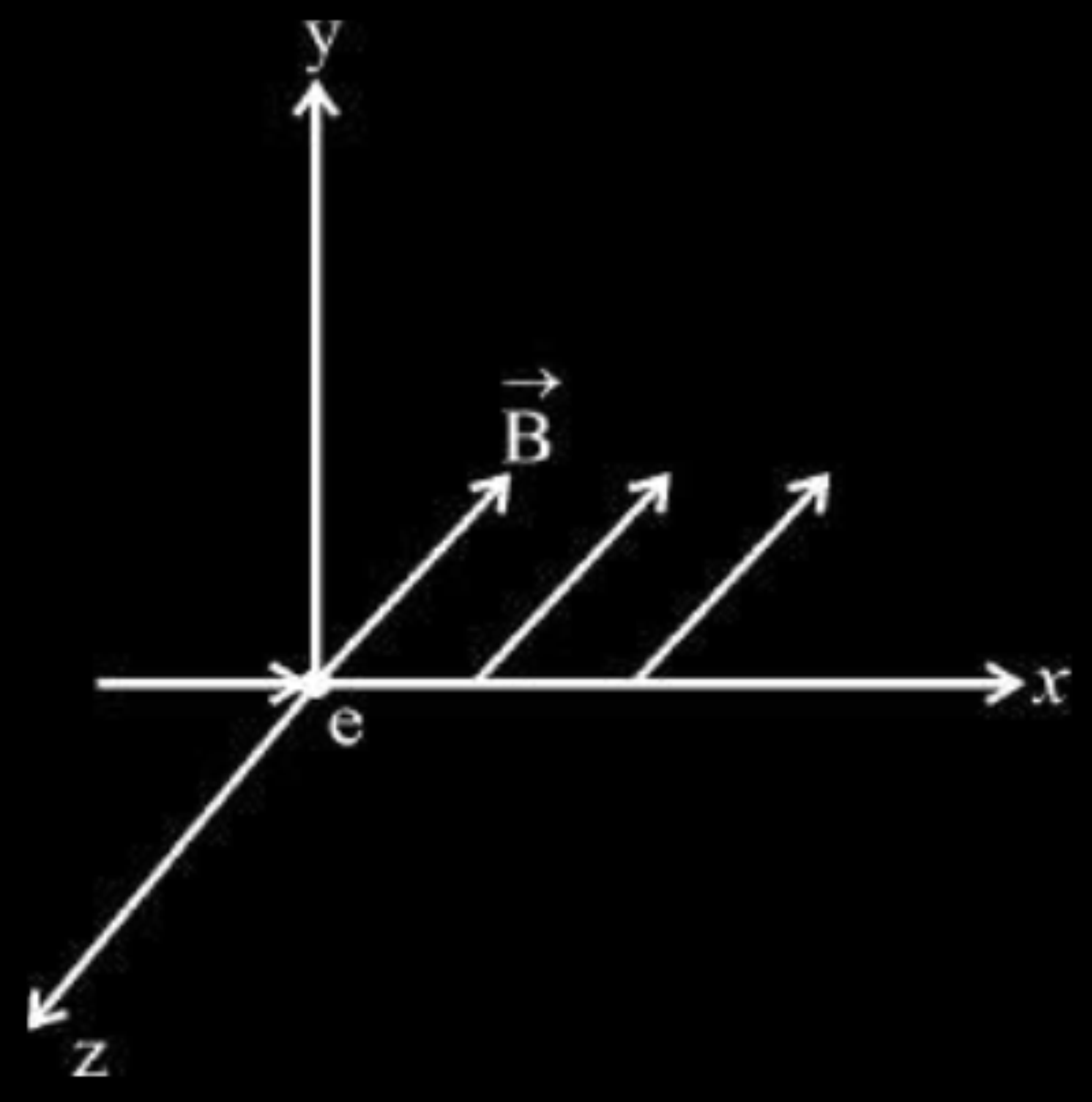
(ii) A current carrying circular loop of area A produces a magnetic field B at its centre. Show that the magnetic moment of the loop is $\dfrac{2\,BA}{\mu_0}\sqrt{\dfrac{A}{\pi}}$.

i) Finding the direction of magnetic field near points P,Q and R	½ + ½ + ½
Conclusion about the relative magnitude of magnetic field.	½ + ½ + ½
ii) Showing the given expression of magnetic moment.	2

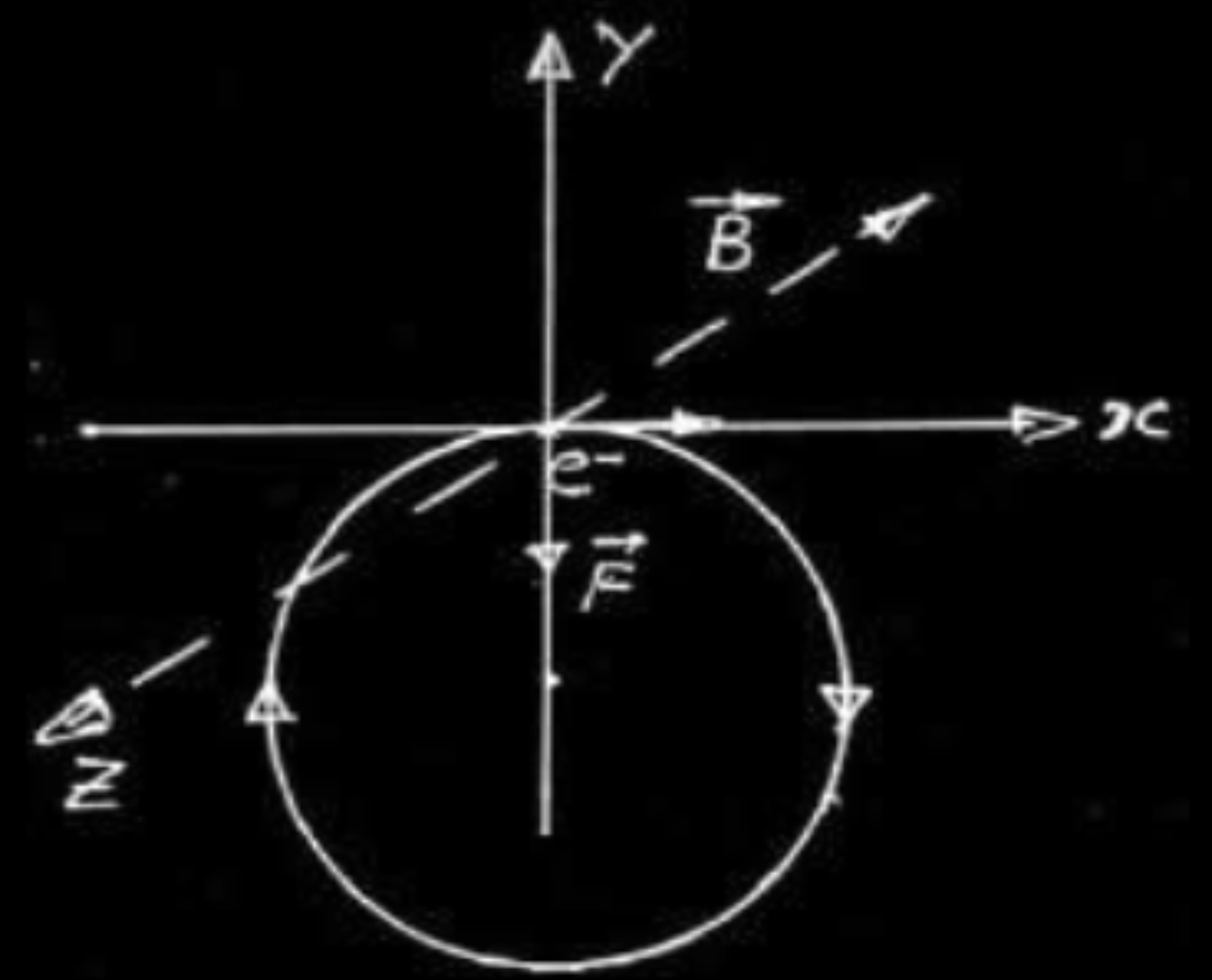
i) <u>Near point P</u>	
Magnetic field is acting into the plane of the paper as Force is acting upwards.	½
<u>Near point Q</u>	
Magnetic field is into the plane of paper as force is acting upwards.	½
<u>Near point R</u>	
Magnetic field is acting out of the plane of the paper as $\vec{F}$ is acting downwards.	½
<u>Relative Magnitude of the Magnetic field.</u>	
As B $\alpha \dfrac{1}{r}$	
Therefore,	
Near point P, magnitude of B is small.	½
Near point Q, B is relatively smaller than point P.	½
Near point R, B is relatively larger than point P.	½
$(B_Q < B_P < B_R)$	
ii) Let r be the radius of the circular coil and I is the current in the coil then	
$B = \dfrac{\mu_0 I}{2r}$ or $I = \dfrac{2Br}{\mu_0}$	½
$A = \pi r^2$ $\quad r = \sqrt{\dfrac{A}{\pi}}$	½
$M = IA$	½
$= \dfrac{2Br}{\mu_0} A$	
$= \dfrac{2BA}{\mu_0}\sqrt{\dfrac{A}{\pi}}$	½



20. An electron moves along $+x$ direction. It enters into a region of uniform magnetic field $\vec{B}$ directed along –z direction as shown in fig. Draw the shape of trajectory followed by the electron after entering the field.



1

 	1
Alternatively: Circular path in the X-Y plane in clockwise sense.	½+½
[**Note:** If the student just writes, force on the electron will be along negative Y axis, i.e. $F = -e(v\ \hat{i}) \times (B(-\hat{k}) = evB(-\hat{j})\ \ award\ {}^{1}\!/_{2}\, mark\ \ only)$	

9. A region has a uniform magnetic field in it. A proton enters into the region with velocity making an angle of 45° with the direction of the magnetic field. In this region the proton will move on a path having the shape of a *1*

(A) straight line

(B) circle

(C) spiral

(D) helix

9. A region has a uniform magnetic field in it. A proton enters into the region with velocity making an angle of 45° with the direction of the magnetic field. In this region the proton will move on a path having the shape of a *1*

(A) straight line

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(C) spiral

(D) helix

(D)

Helix

3. An electron is released from rest in a region of uniform electric and magnetic fields acting parallel to each other. The electron will *1*

(A) move in a straight line.

(B) move in a circle.

(C) remain stationary.

(D) move in a helical path.

3. An electron is released from rest in a region of uniform electric and magnetic fields acting parallel to each other. The electron will *1*

(A) move in a straight line.

(B) move in a circle.

(C) remain stationary.

(D) move in a helical path.

(A) move in a straight line.

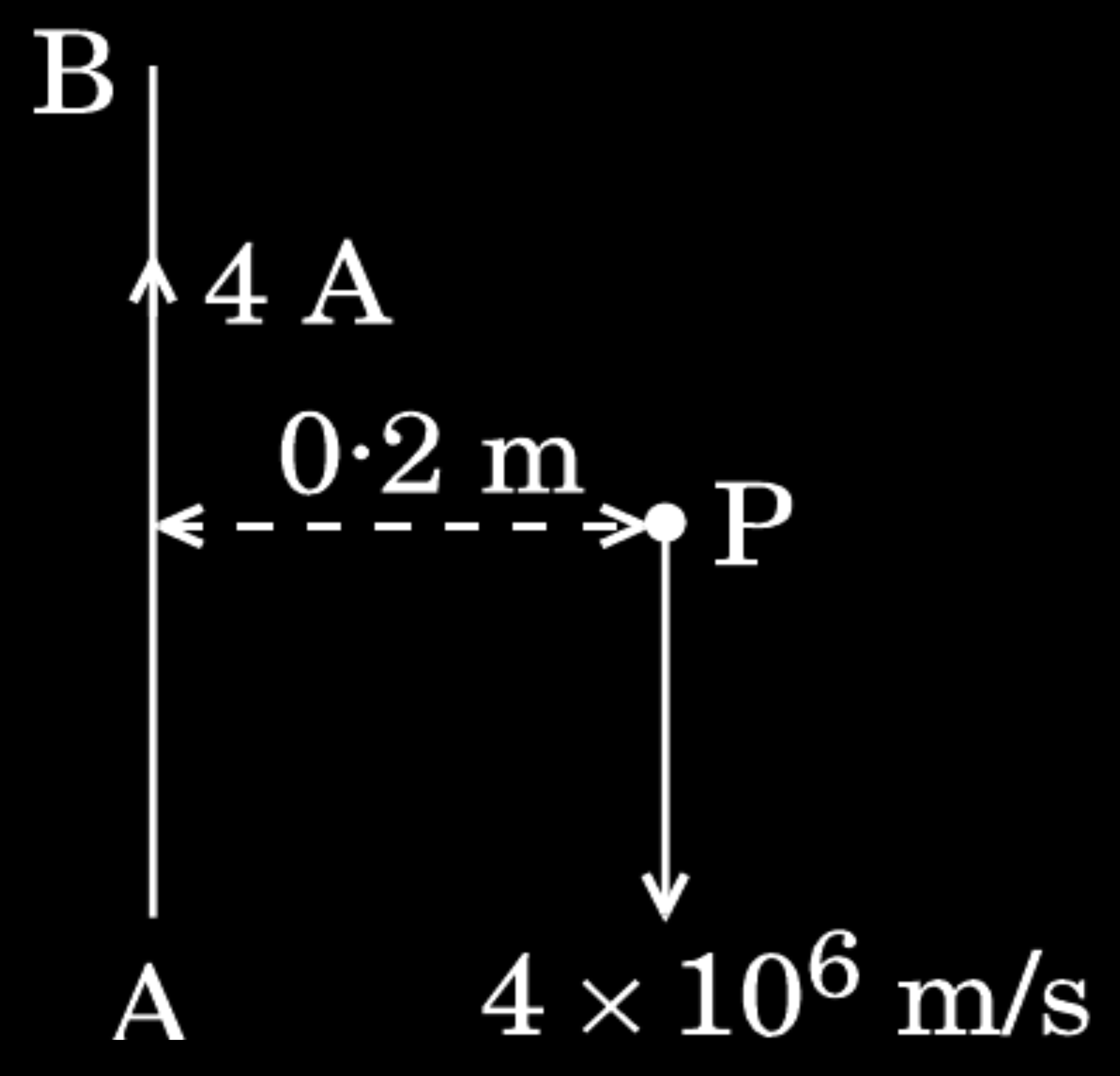
(a) Obtain the conditions under which an electron does not suffer any deflection while passing through a magnetic field.

(b) Two protons P and Q moving with the same speed pass through the magnetic fields $\vec{B}_1$ and $\vec{B}_2$ respectively, at right angles to the field directions. If $|\vec{B}_2| > |\vec{B}_1|$, which of the two protons will describe the circular path of smaller radius ? Explain.

a) Condition for no deflection -	1
b) Conclusion for greater radius -	1

(a) No deflection if electron moves parallel or anti parallel to the magnetic field	1
(b)	
$r = \dfrac{mv}{Bq}$	½
$\dfrac{r_1}{r_2} = \dfrac{B_2}{B_1}$	½
$As \quad B_1 \langle B_2 \qquad r_2 \langle r_1$	
$Alternatively$	
$[r = \dfrac{l}{B}$	½
r_2 *is smaller because* $B]$	½

A long straight wire AB carries a current of 4 A. A proton P travels at 4×10^6 ms^{-1} parallel to the wire 0·2 m from it and in a direction opposite to the current as shown in the figure. Calculate the force which the magnetic field due to the current carrying wire exerts on the proton. Also specify its direction.



i)	Formula for Magnetic field	½
ii)	Formula for force	½
iii)	Calculation & Results	½
iv)	Direction of force	½

Magnetic field at Point P due current carrying straight wire AB $B = \frac{\mu_o}{2\pi}\frac{I}{r}$	½
Force acting on the moving proton in the magnetic field $F = Bqv\sin\theta$	½
Therefore $F = \frac{\mu_o}{2\pi}\frac{I}{r} \times qv\sin\theta$ $= \frac{2 \times 10^{-7} \times 4 \times 1.6 \times 10^{-19} \times 4 \times 10^{6} \sin 90}{0.2}$	
$= 2.56 \times 10^{-18}$ N	½
Direction of force at point P is towards right. (away from AB)	½

A proton, a deuteron and an alpha particle, are accelerated through the same potential difference and then subjected to a uniform magnetic field $\vec{B}$, perpendicular to the direction of their motions. Compare (i) their kinetic energies, and (ii) if the radius of the circular path described by proton is 5 cm, determine the radii of the paths described by deuteron and alpha particle.

i)	Relation between K.E. and accelerating potential	½
ii)	Comparison of Kinetic Energies	1
iii)	Determination of radii of path of deuteron and alpha particle	1½

A proton is accelerated through a potential difference V, subjected to a uniform magnetic field acting normal to the velocity of the proton. If the potential difference is doubled, how will the radius of the circular path described by the proton in the magnetic field change ?

• For writing the expression for radius 1/2 • To find the change in the radius of the circular orbit 1/2 $r = \frac{1}{B}\sqrt{\frac{2mV}{q}}$ $r^{'} = \sqrt{2}r$ $Alternatively\ r\alpha\sqrt{V}$ $\frac{r'}{r} = \sqrt{2}$ $Even\ if\ student\ writes\ Bq\text{v} = \frac{m\text{v}^2}{r}\ award\ one\ mark$	½ ½ ½ ½

An α-particle and a proton of the same kinetic energy are in turn allowed to pass through a magnetic field $\vec{B}$, acting normal to the direction of motion of the particles. Calculate the ratio of radii of the circular paths described by them.

Formula	½
Calculation of ratio of radii	1½

radius r = $\frac{mv}{q\ B}$ = $\frac{\sqrt{2mk}}{qB}$	
$K_\alpha = K_{proton}$	
$M_\alpha = 4\ m_p$	½
$q_\alpha = 2q_p$	
$\frac{r_\alpha}{r_p} = \frac{\frac{\sqrt{2m_\alpha K}}{q_\alpha B}}{\frac{\sqrt{2m_p K}}{q_p B}}$	½
$= \sqrt{\frac{m_\alpha}{m_p}} \times \sqrt{\frac{q_p}{q_\alpha}}$	½
$= \sqrt{4} \times ½ = 1$	

CBSE 2018

Two protons of equal kinetic energies enter a region of uniform magnetic field. The first proton enters normal to the field direction while the second enters at 30° to the field direction. Name the trajectories followed by them.

CBSE 2017

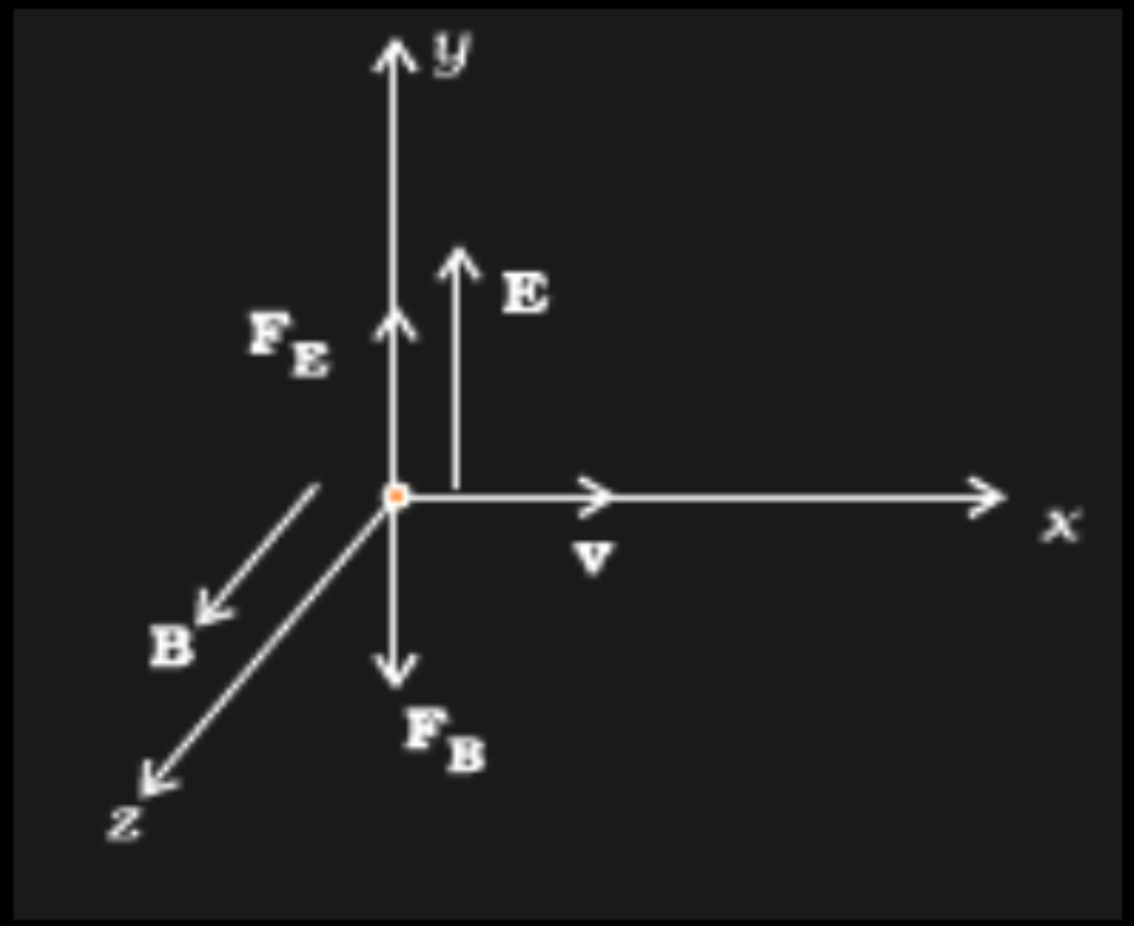
Find the condition under which the charged particles moving with different speeds in the presence of electric and magnetic field vectors can be used to select charged particles of a particular speed.

Condition		
i.	For directions of $\vec{E}, \vec{B}, \vec{v}$	1
ii.	For magnitudes of $\vec{E}, \vec{B}, \vec{v}$	1

(i) The velocity $\vec{v}$, of the charged particles, and the $\vec{E}$ and $\vec{B}$ vectors, should be mutually perpendicular. ½

Also the forces on q, due to $\vec{E}$ and $\vec{B}$, must be oppositely directed. ½

(Also accept if the student draws a diagram to show the directions.)



(ii) $qE = qvB$ ½

$or\ v = \dfrac{E}{B}$ ½

[Alternatively, The student may write:

Force due to electric field = $q\vec{E}$ ½

Force due to magnetic field = $q\ (\vec{v} \times \vec{B})$ ½

The required condition is

$$q\vec{E} = -q\ (\vec{v} \times \vec{B})$$ ½

$$\left[or\ \vec{E} = -(\vec{v} \times \vec{B}) = (\vec{B} \times \vec{v})\right]$$ ½

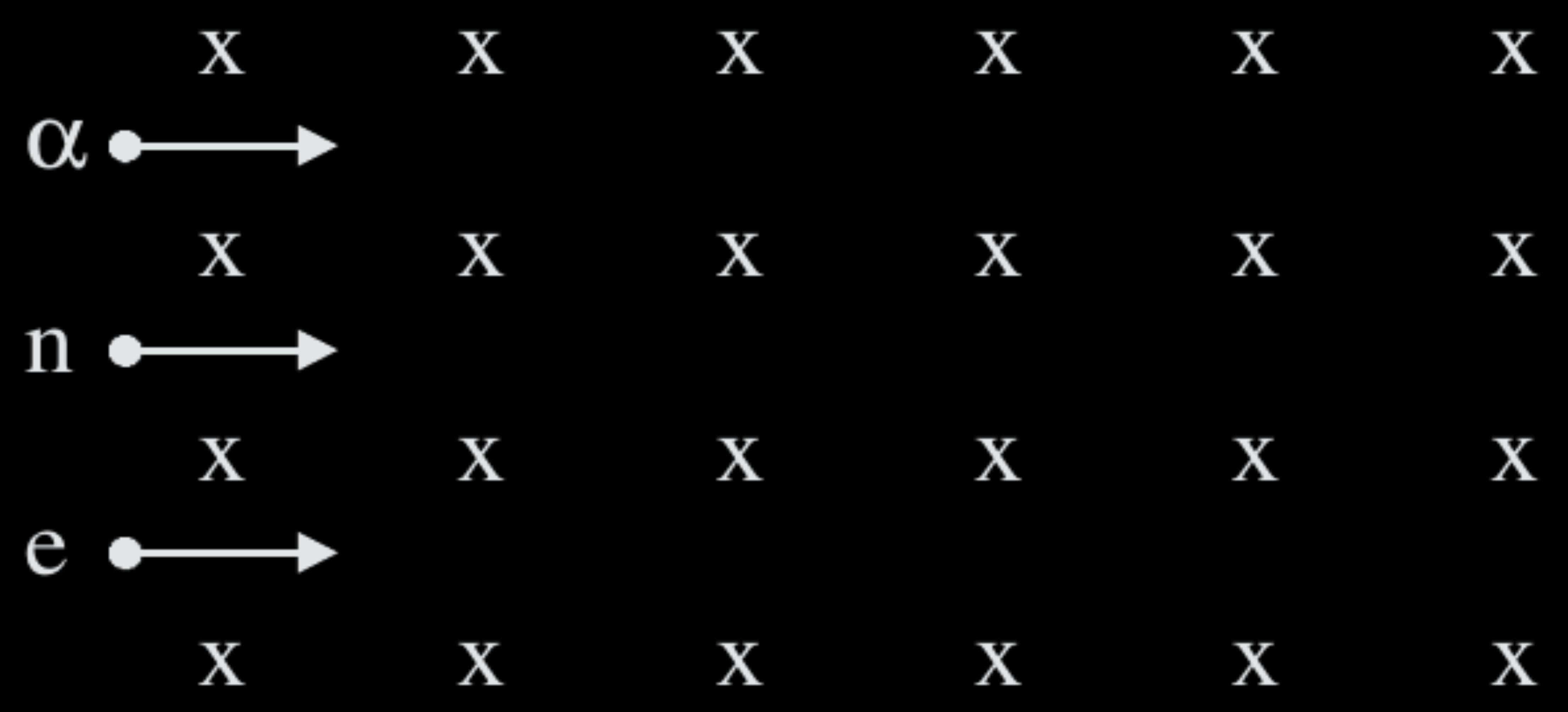
(Note: Award 1 mark only if the student just writes:

"The forces, on the charged particle, due to the electric and magnetic fields, must be equal and opposite to each other")]

CBSE 2016

(a) Write the expression for the magnetic force acting on a charged particle moving with velocity v in the presence of magnetic field B.

(b) A neutron, an electron and an alpha particle moving with equal velocities, enter a uniform magnetic field going into the plane of the paper as shown. Trace their paths in the field and justify your answer.



a)	Expression for the magnetic force	1
b)	Trace of paths	½ + ½ + ½
	Justification	½

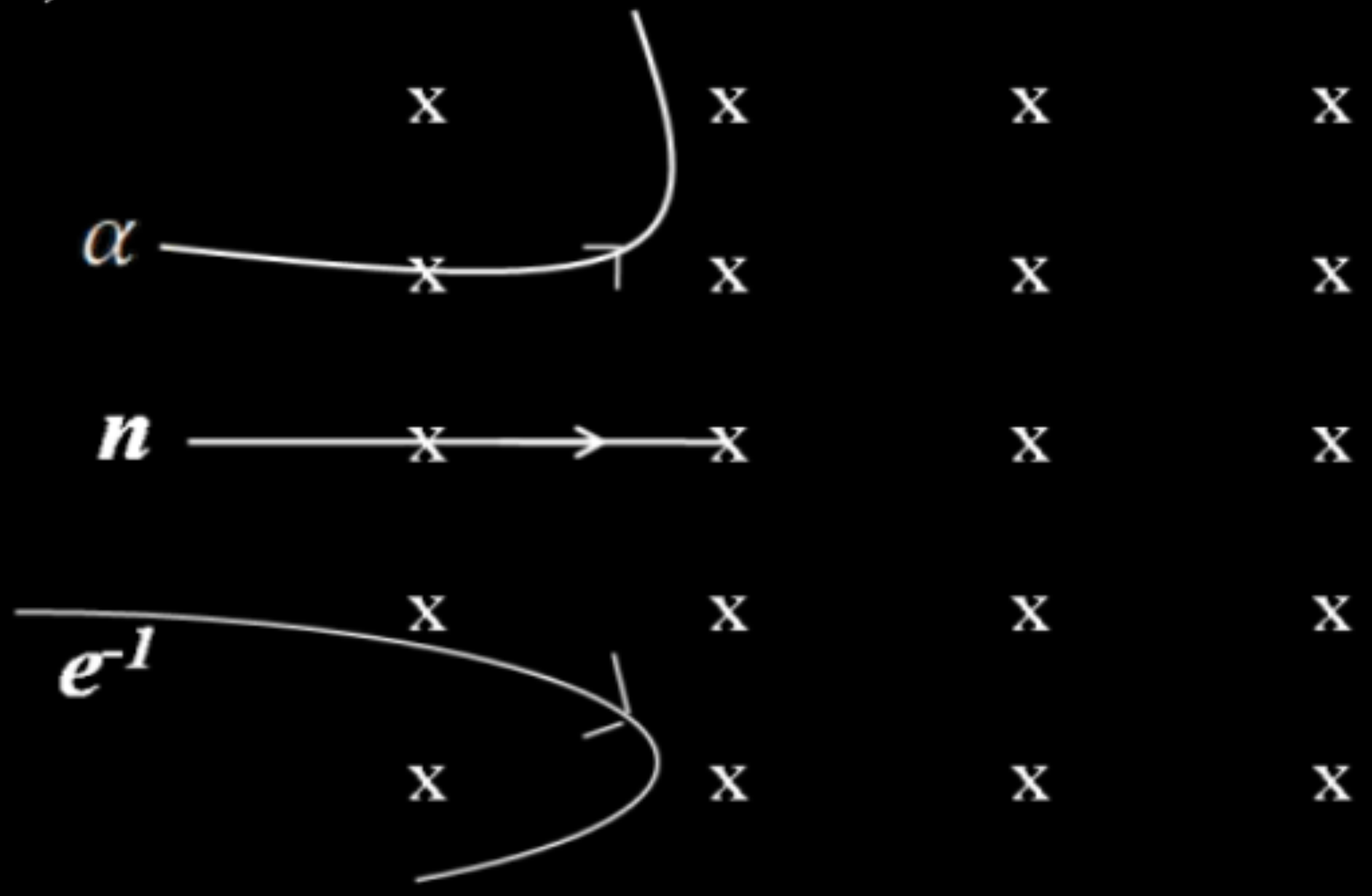
$\vec{F} = q\,(\vec{v} \times \vec{B})$ 1

(Give Full credit of this part even if a student writes:

$F = qvB\ \text{Sin}\theta$ and

Force (F) acts perpendicular to the plane containing $\vec{v}$ and $\vec{B}$)

b)



½ +

½+

½

Justification: Direction of force experienced by the particle will be according to the Fleming's Left hand rule / (any other alternative correct rule.) ½

(a) Consider a beam of charged particles moving with varying speeds. Show how crossed electric and magnetic fields can be used to select charged particles of a particular velocity ?

(b) Name another device/machine which uses crossed electric and magnetic fields. What does this machine do and what are the functions of magnetic and electric fields in this machine ? Where do these field exist in this machine ? Write about their natures.

a)	Velocity selection condition	1
b)	Name of device	½
	What does the machine do	½
	Use of two fields	½ + ½
	Regions of existence of field	½ + ½
	Nature of fields	½ + ½

a) $qE = Bqv$	
$v = E/B$	1
(b) Name of the device: Cyclotron	
It accelerates charged particles/ions	½
Electric field accelerates the charged particles.	½
Magnetic field makes particles to move in circle.	1
Electric field exists between the Dees.	
Magnetic field exists both inside and outside the dees.	1
Magnetic field is uniform / constant.	
Electric field is oscillating/ alternating in nature.	1

What can be the cause of helical motion of a charged particle ?

What can be the cause of helical motion of a charged particle ?

$\boxed{\begin{array}{l}\text{Charged particle moves inclined to the magnetic field} \\ \text{(angle between } \vec{\vartheta} \text{ and } \vec{B} \text{ is neither } \pi/2 \text{ nor } 0) \\ \text{(component of } \vec{\vartheta}\text{, parallel to } \vec{B}\text{, is not zero.)}\end{array}}$

(iii) An electron after being accelerated through a potential difference of 100 V enters a uniform magnetic field of 0.004 T perpendicular to its direction of motion. Calculate the radius of the path described by the electron.

(iii) An electron after being accelerated through a potential difference of 100 V enters a uniform magnetic field of 0.004 T perpendicular to its direction of motion. Calculate the radius of the path described by the electron.

(ii) $r = \frac{mv}{qB} = \frac{\sqrt{2mqV}}{qB}$	½
$r = \frac{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 100}}{1.6 \times 10^{-19} \times 0.004}$ m	½
$r = \frac{5.4 \times 10^{-24}}{6.4 \times 10^{-22}}$ m	½
$r = 8.4 \times 10^{-3}$ m	½